

My QUANTUM COMPUTING JOURNEY

Anna Phan

Monash Qiskit Spring Fest

29 October 2025

ASTRONOMY & ASTROPHYSICS

1

PARTICLE PHYSICS

2

IBM RESEARCH

3

IBM QUANTUM

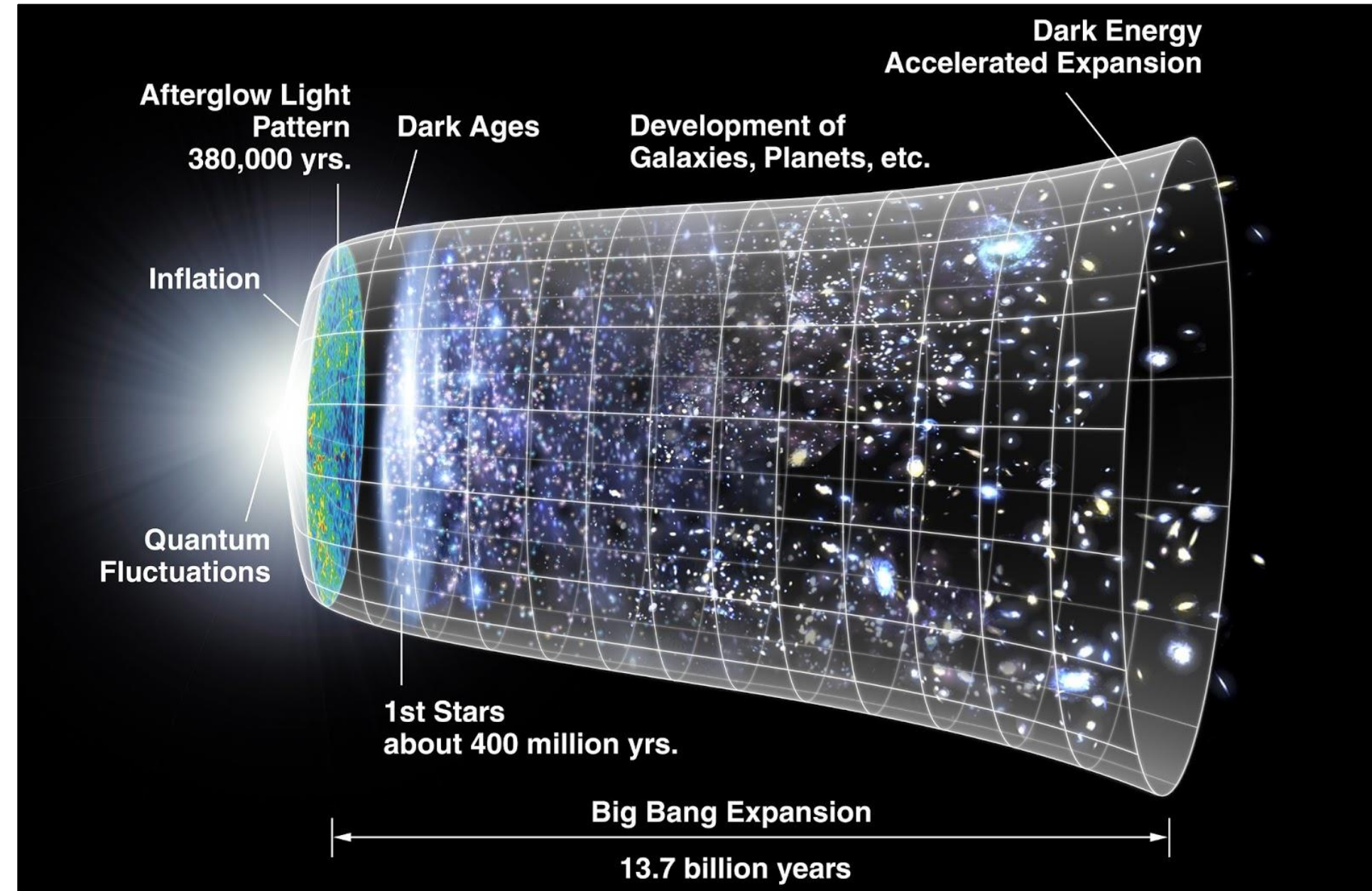
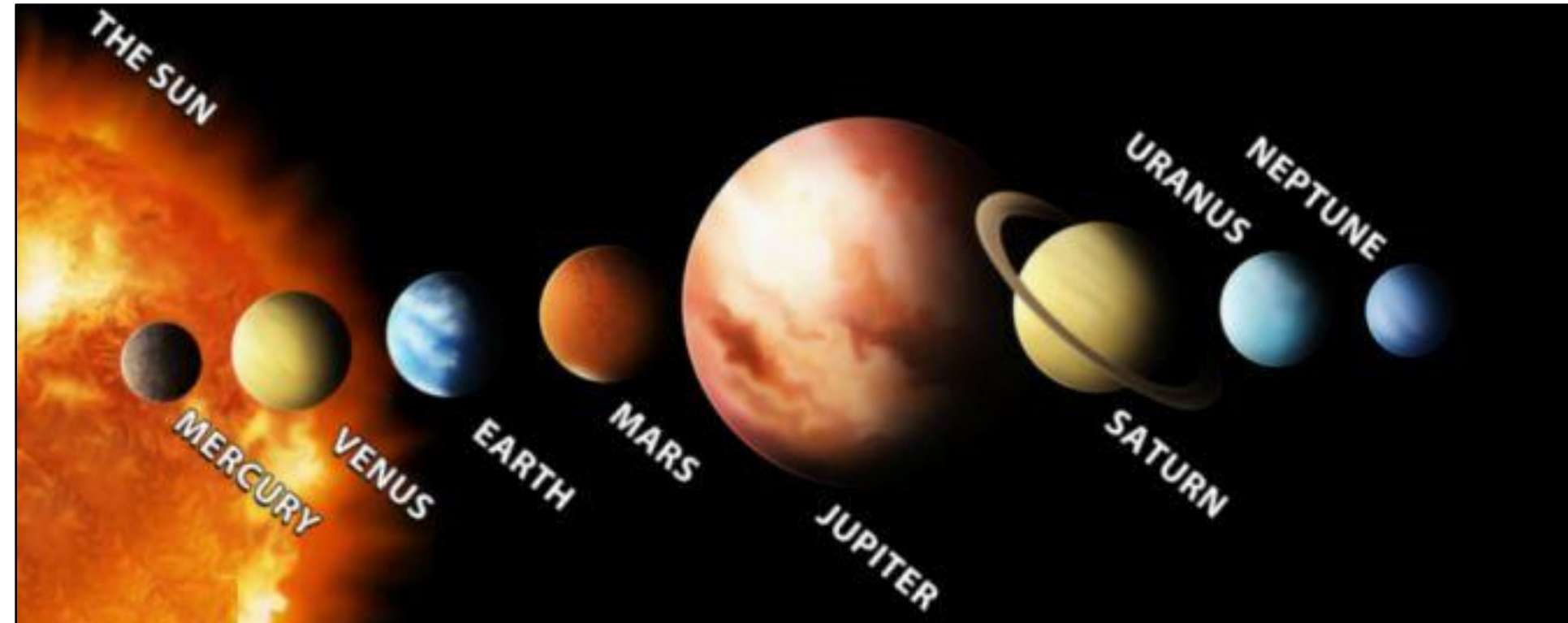
4

The background of the entire image is a deep space photograph showing a vast field of galaxies. These galaxies appear as small, distant points of light in various colors (blue, orange, red, white) against the blackness of space. Some show distinct spiral or elliptical structures. A horizontal grey banner with a slight gradient is positioned across the middle of the image, serving as a background for the title text.

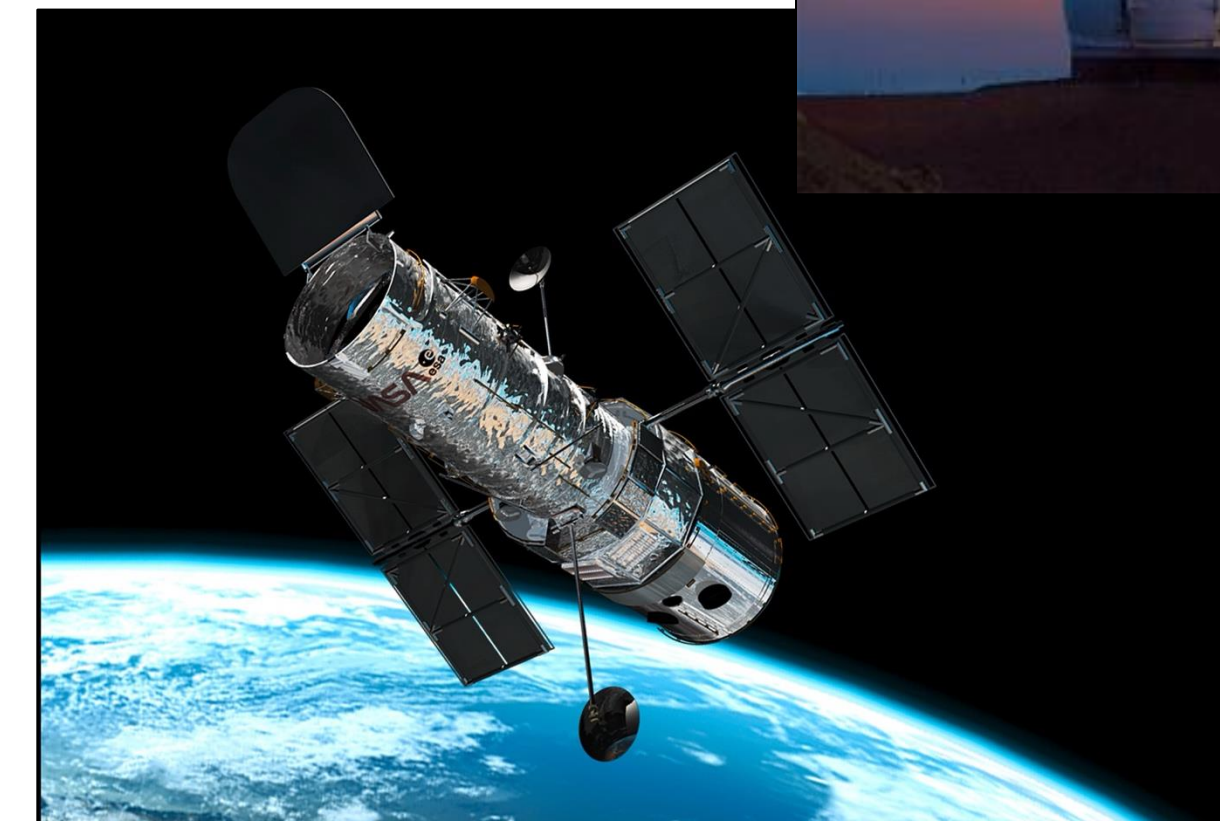
ASTRONOMY & ASTROPHYSICS

WHAT IS ASTROPHYSICS

- Study of the physical properties of celestial objects and their interactions



- Using telescopes



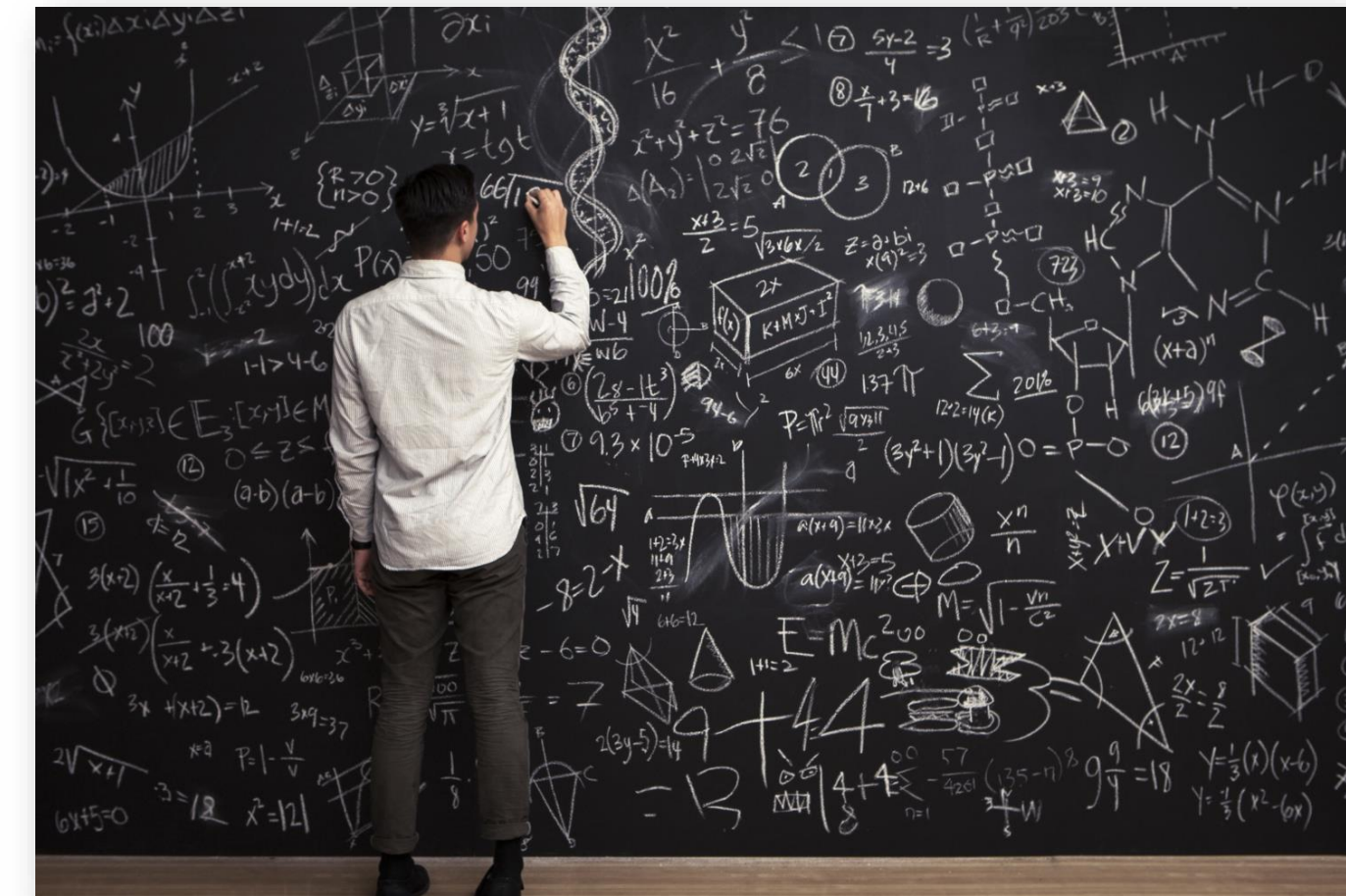
WHAT I LEARNT FROM ASTROPHYSICS

NOT

Telescopes

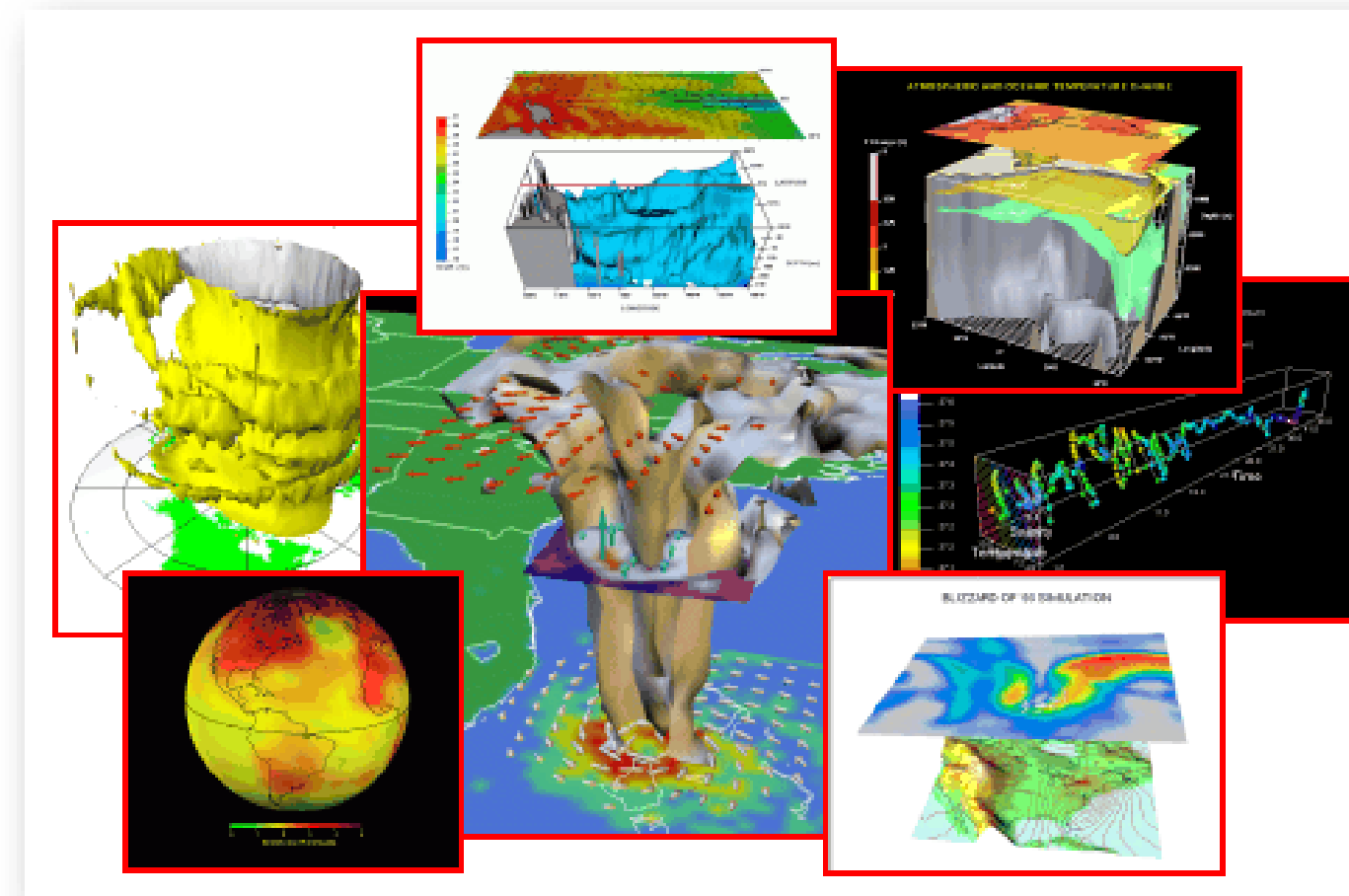


Blackboards

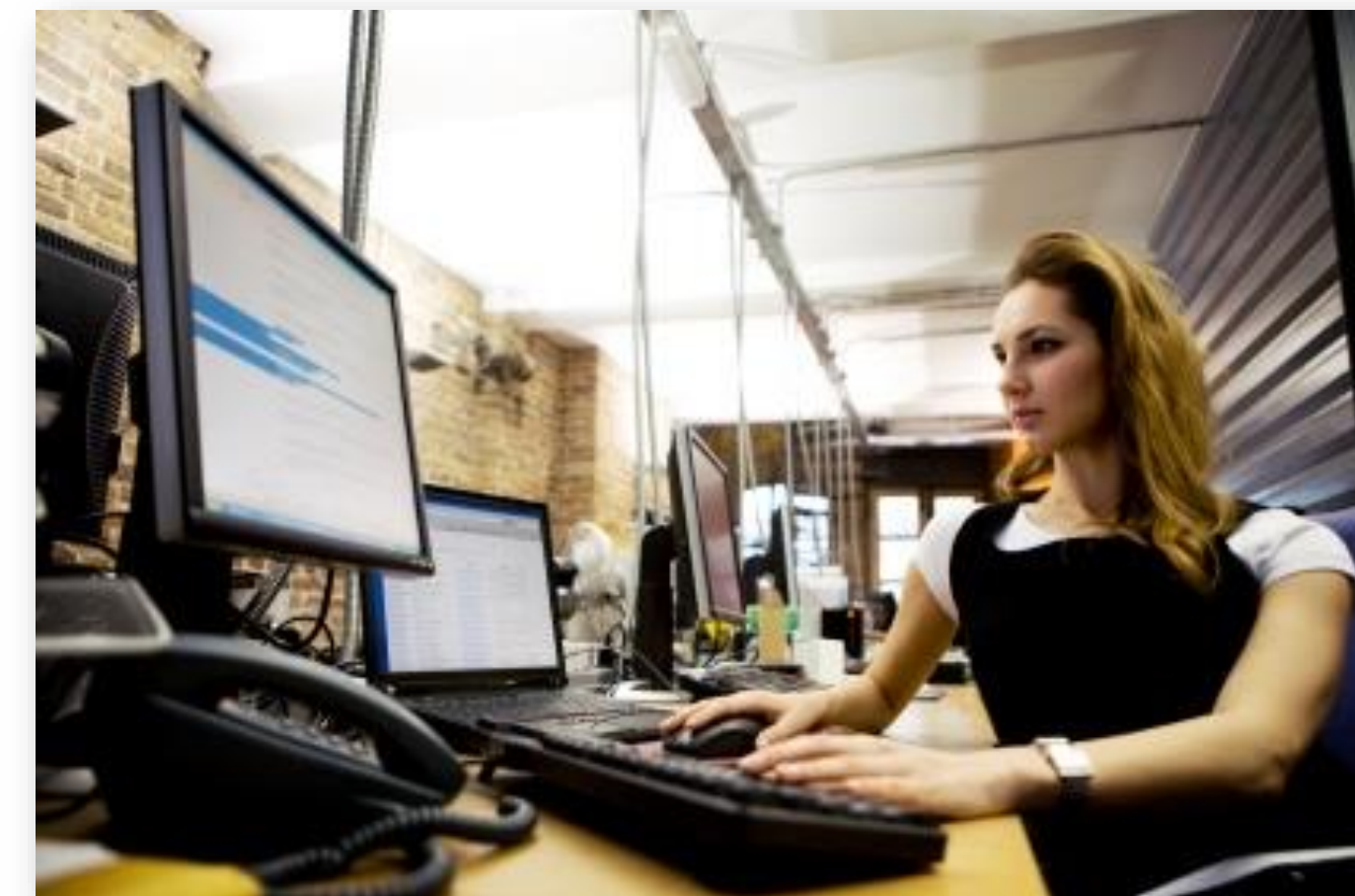


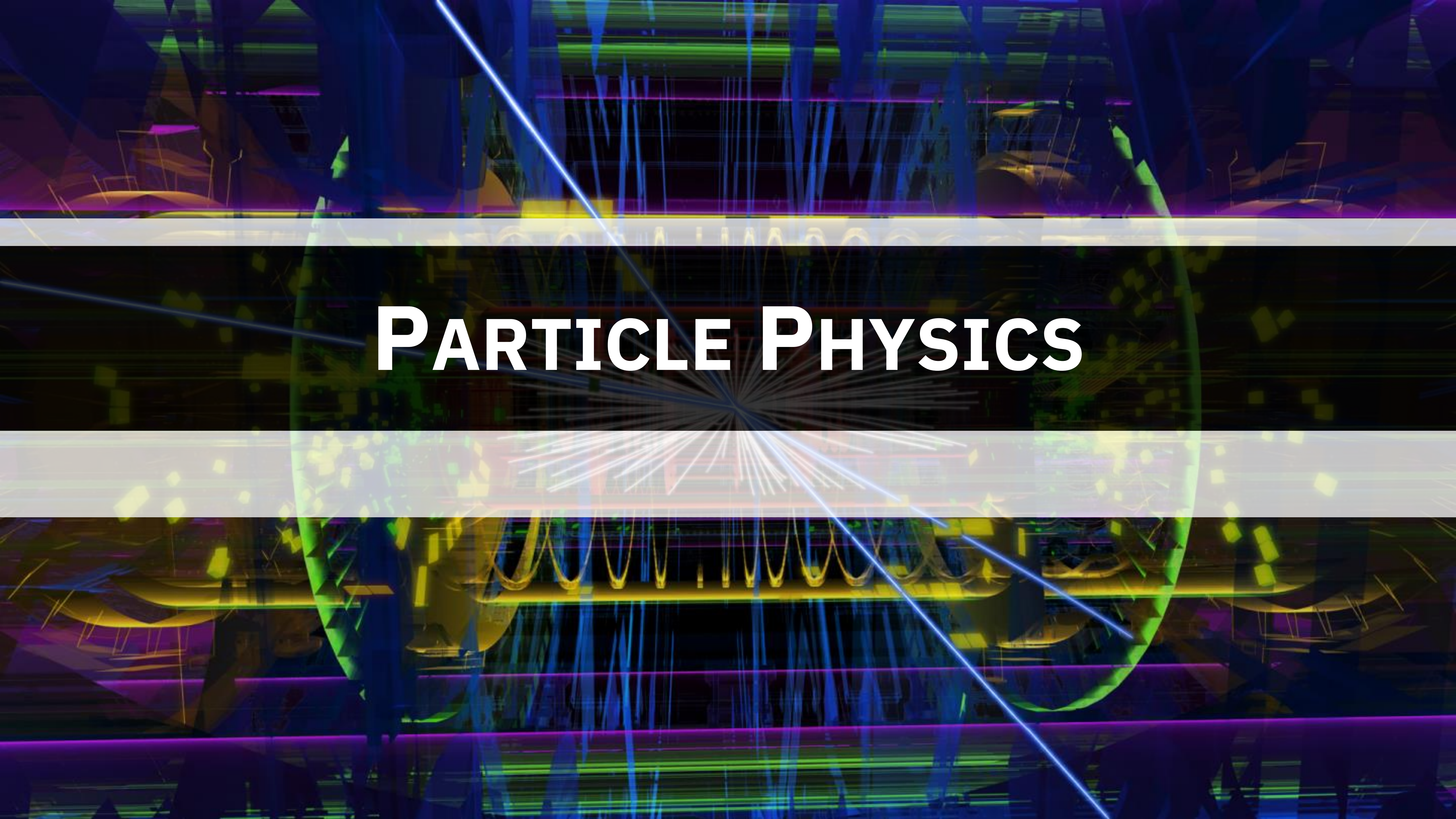
MORE

Graphs



Computers

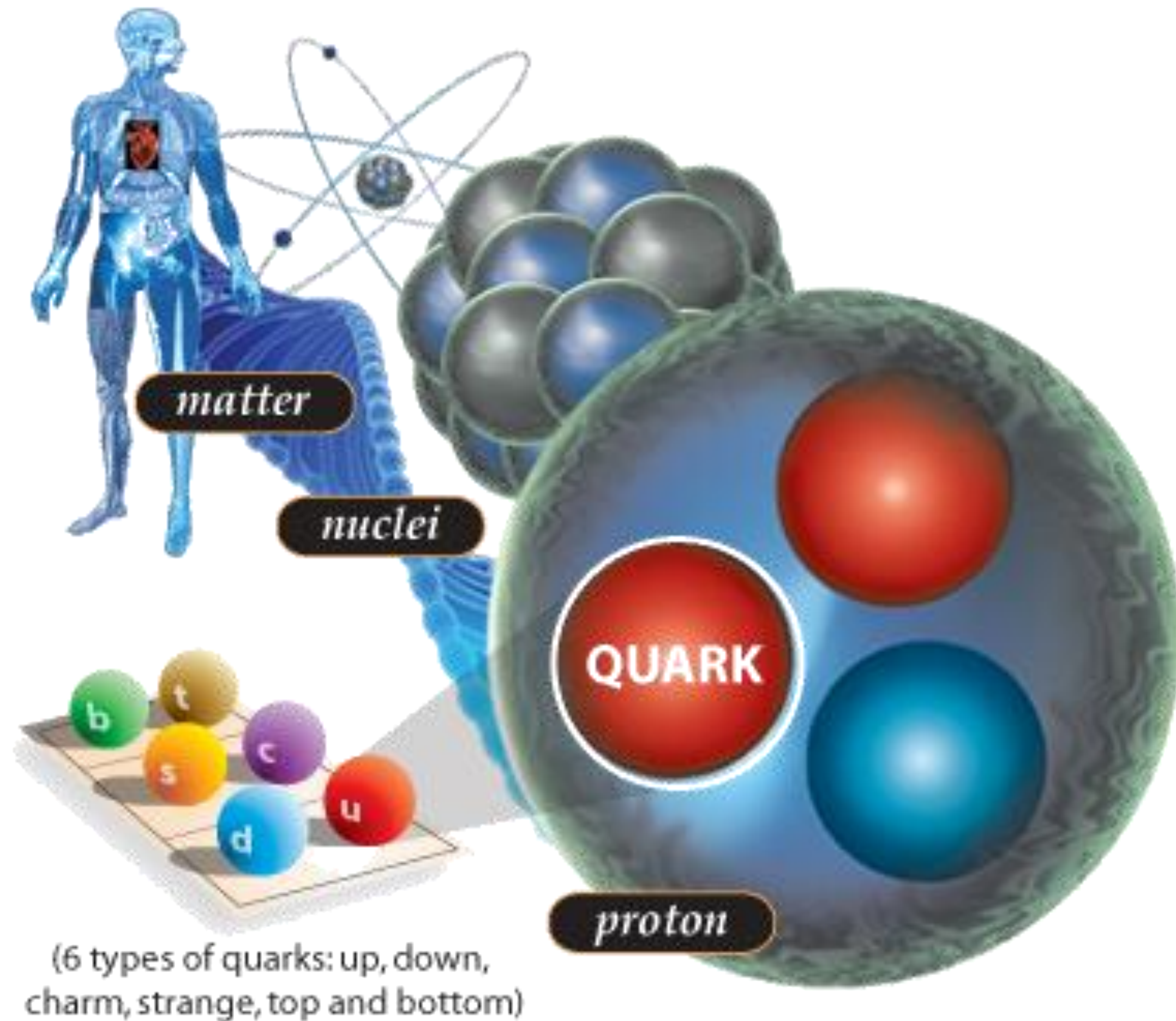


The background is a complex, abstract visualization of particle physics. It features a central point from which numerous thin, white lines radiate outwards, resembling particle tracks or a detector's readout. The background is filled with various geometric shapes and colors, including deep blues, purples, and greens. There are also horizontal bands of light and dark, suggesting different layers or components of a particle accelerator or detector. The overall effect is one of high energy and scientific precision.

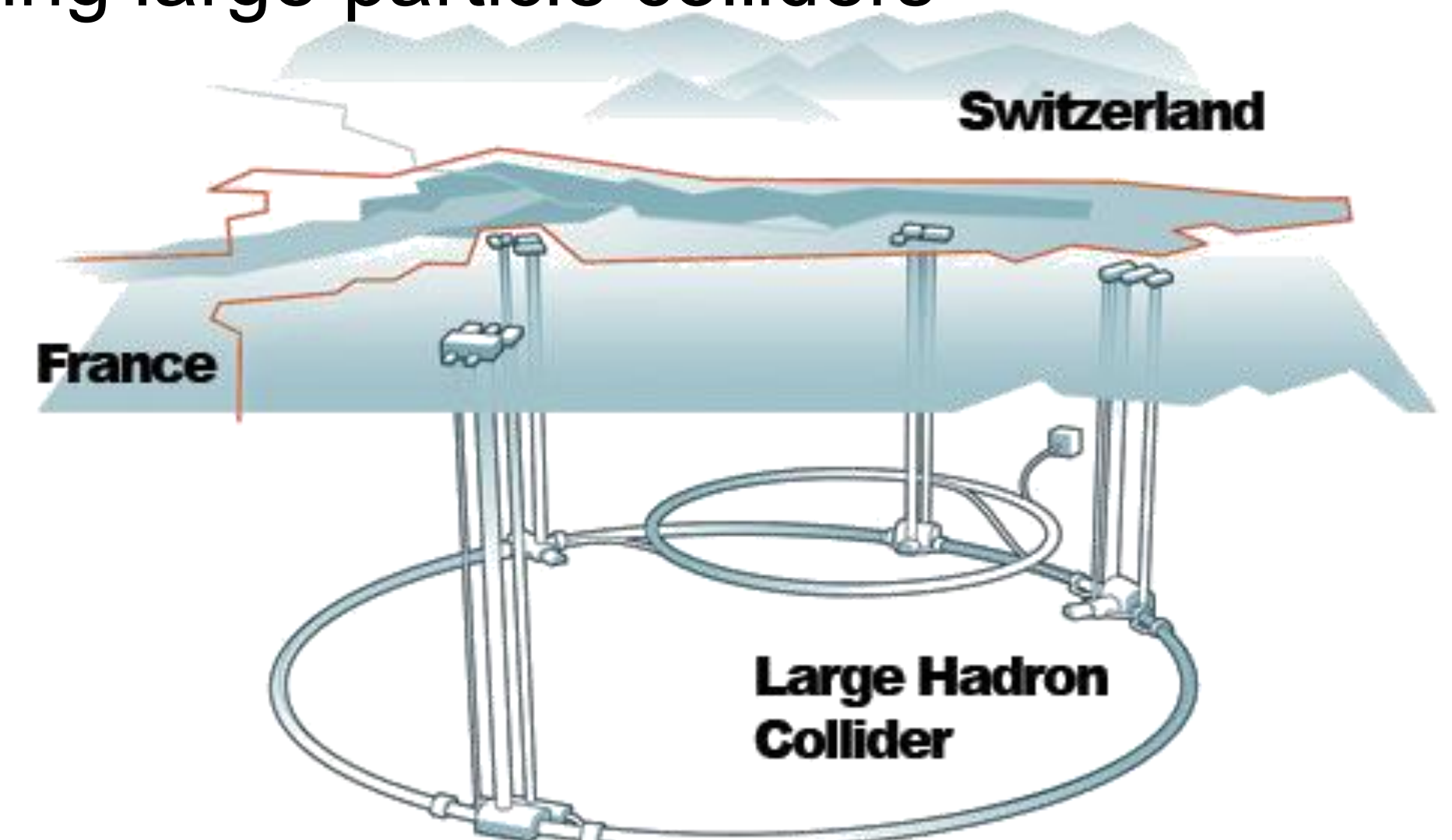
PARTICLE PHYSICS

WHAT IS PARTICLE PHYSICS

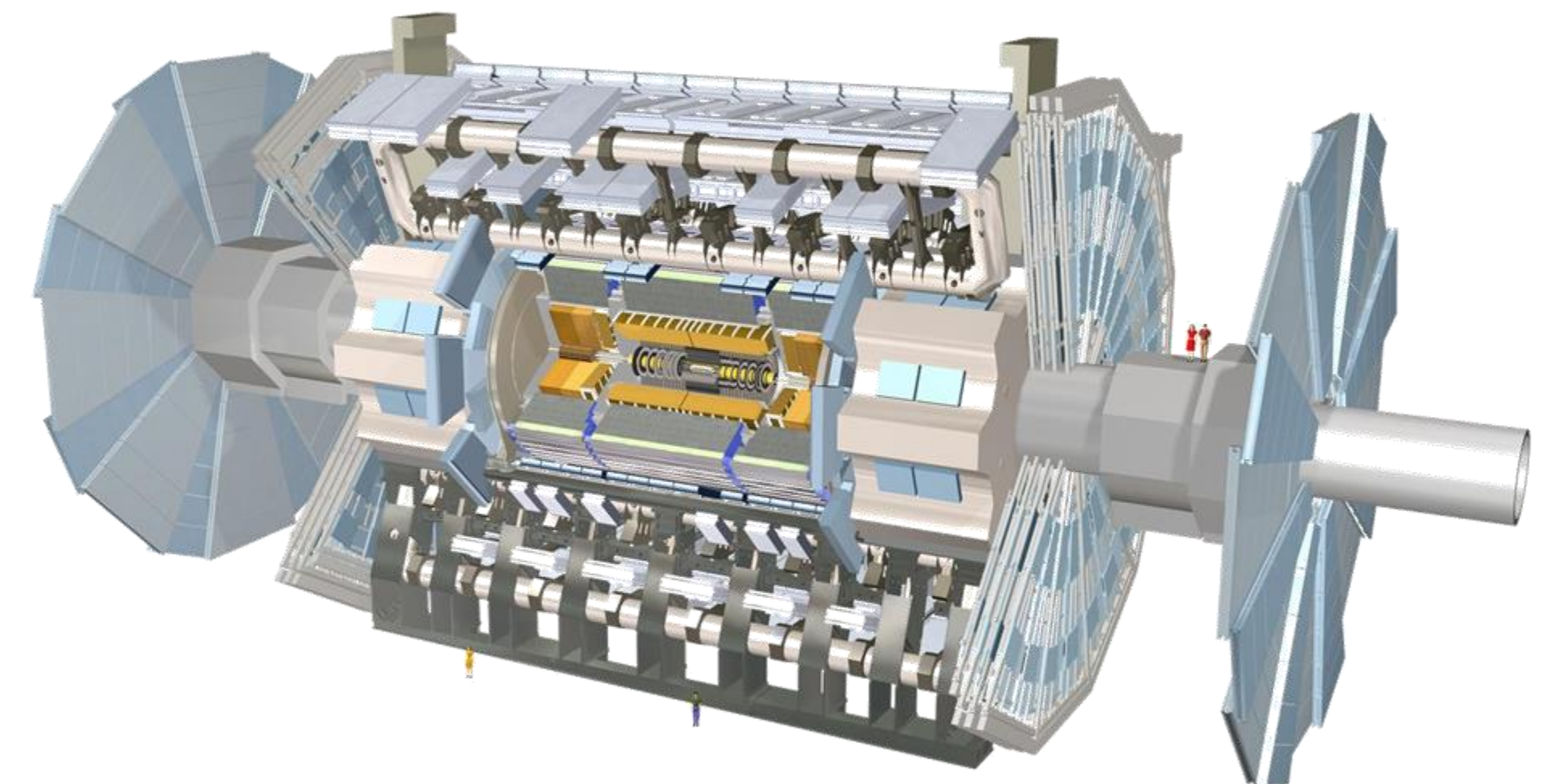
- Study of the basic elements of matter and the forces between them



- Using large particle colliders

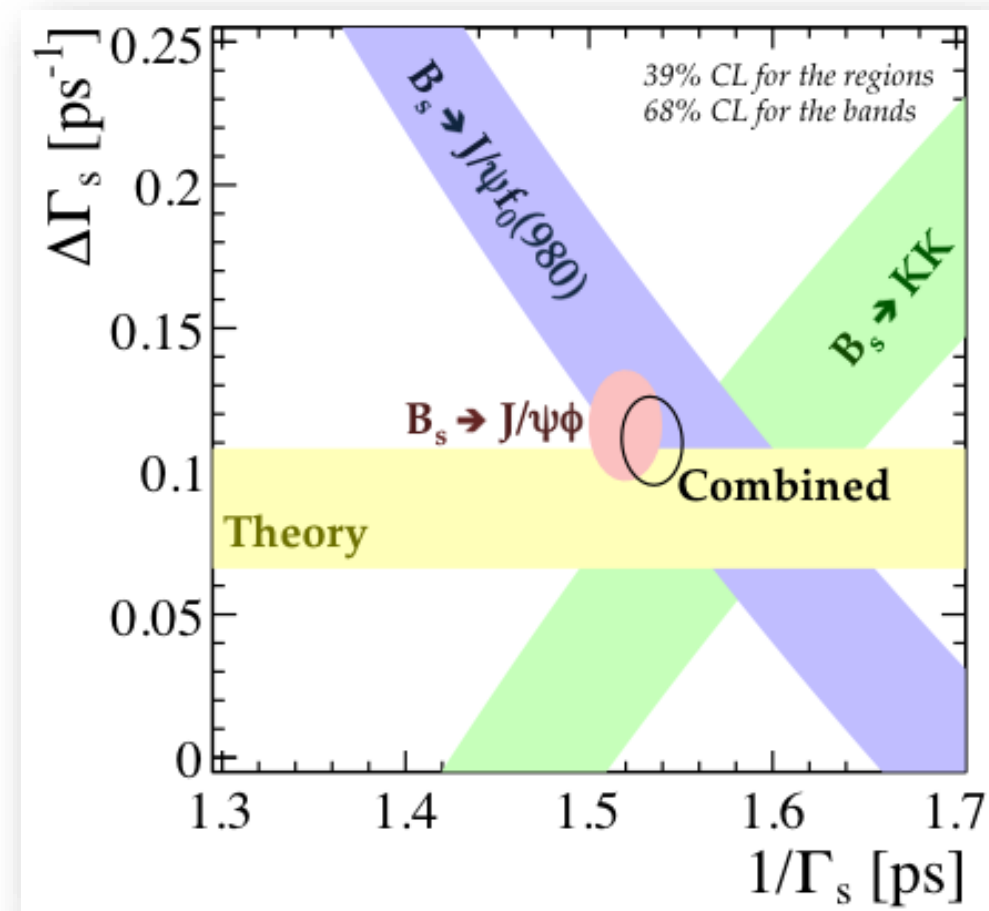


and large particle detectors

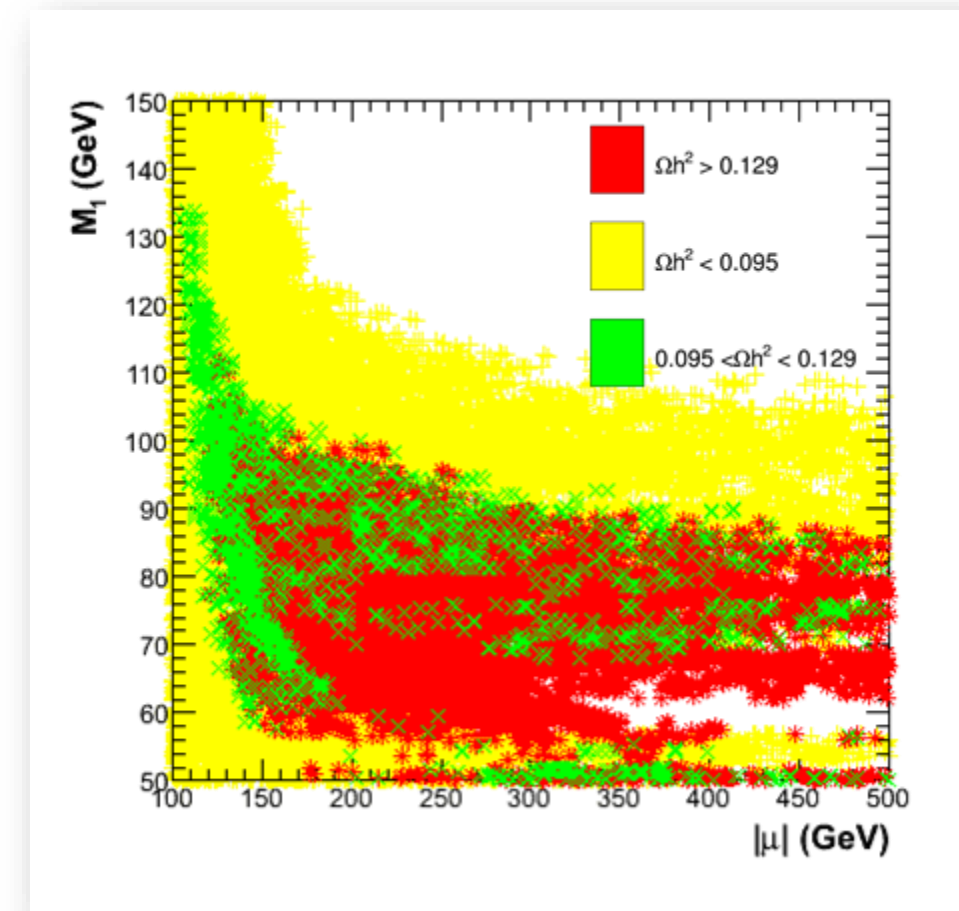


WHAT I LEARNT FROM PARTICLE PHYSICS

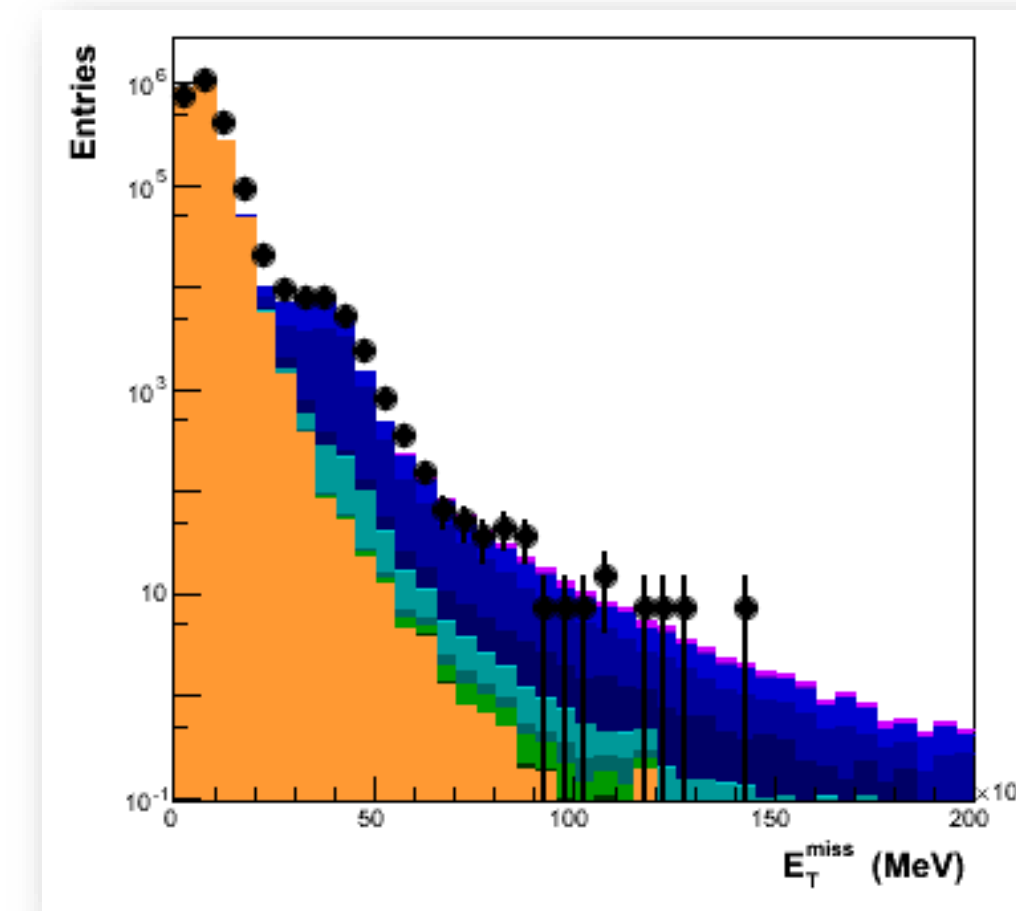
Data Modelling



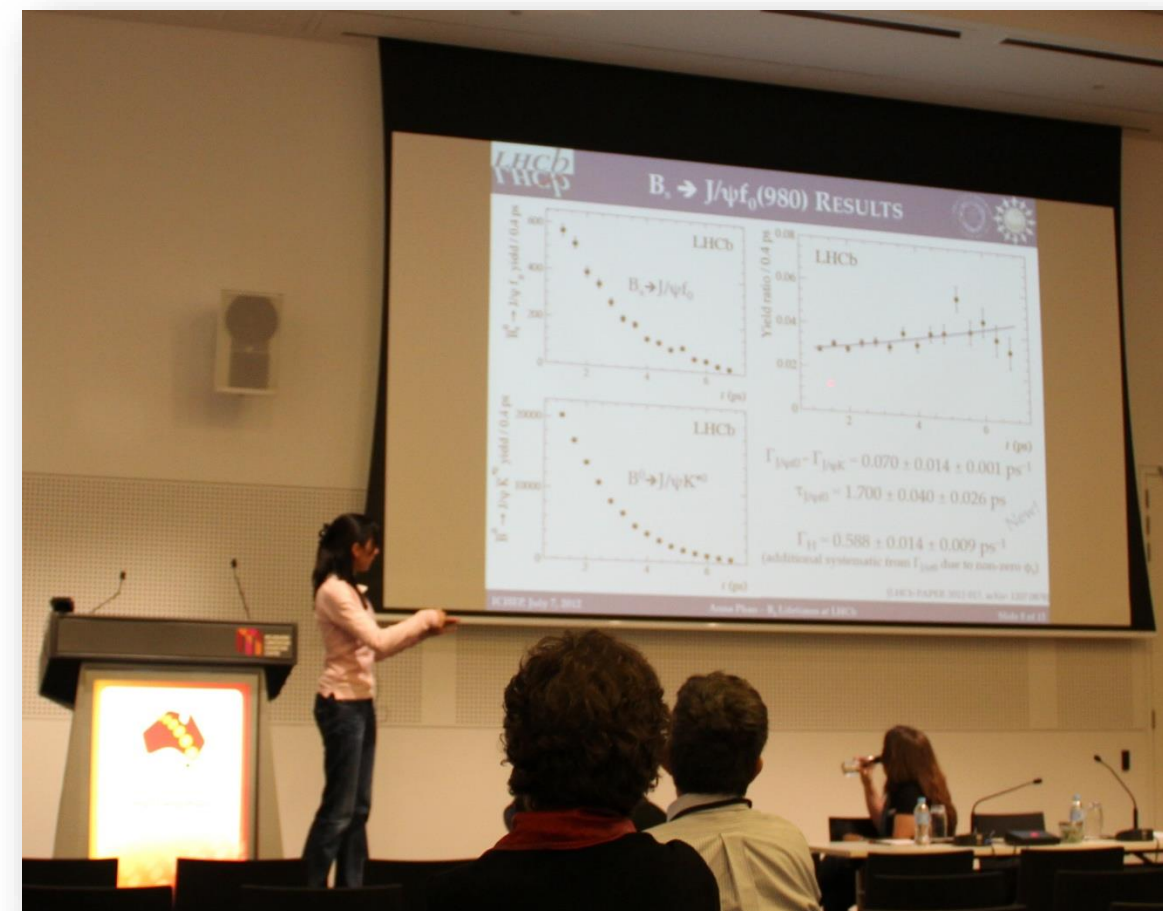
Data Simulation



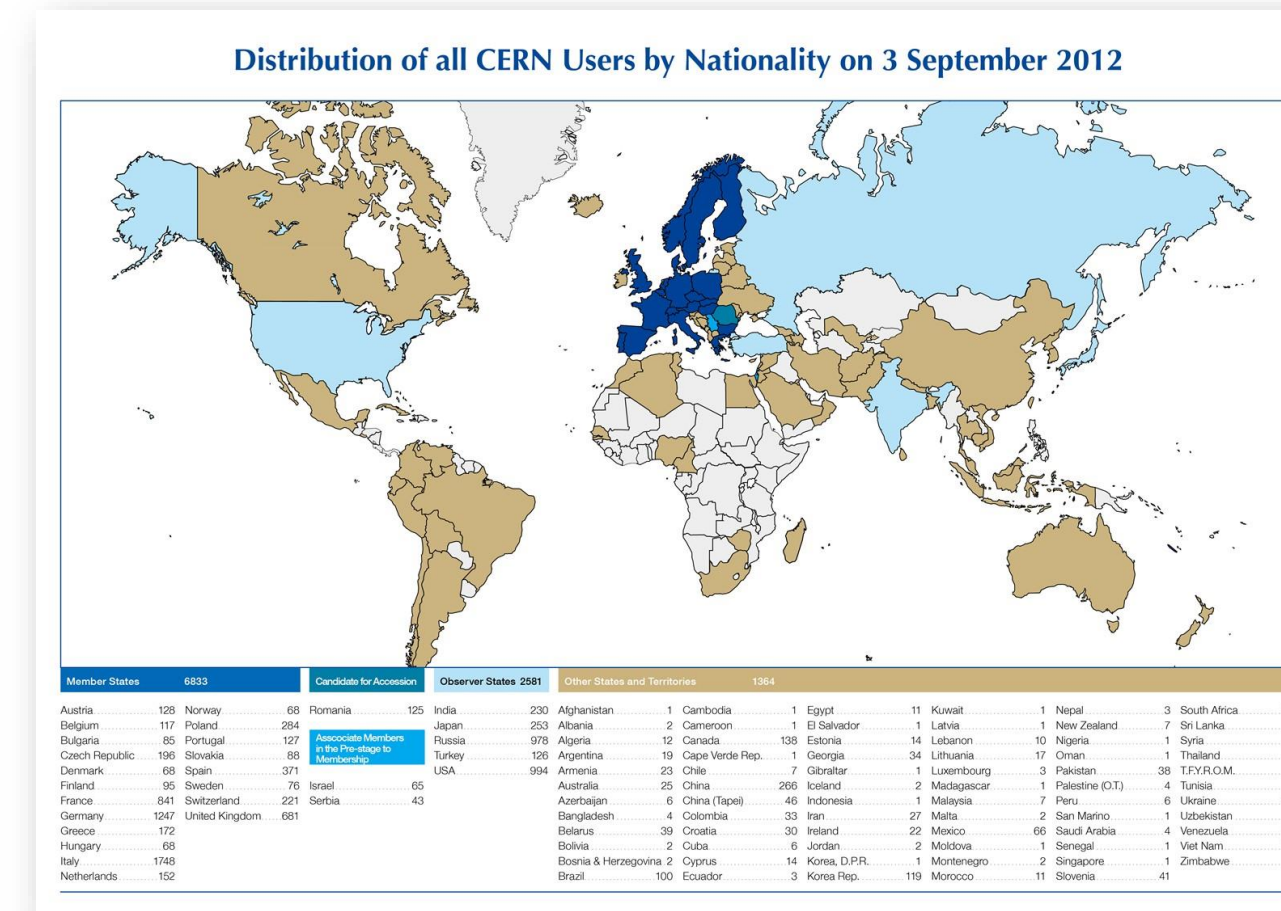
Data Analysis



Communication



Collaboration



IBM RESEARCH

A photograph of a modern, multi-story building with large glass windows and stone accents, illuminated at night. The building is surrounded by a green lawn and trees. The text "IBM RESEARCH" is overlaid in the center.

WHAT IS IBM RESEARCH

Inventing What's Next in Computing

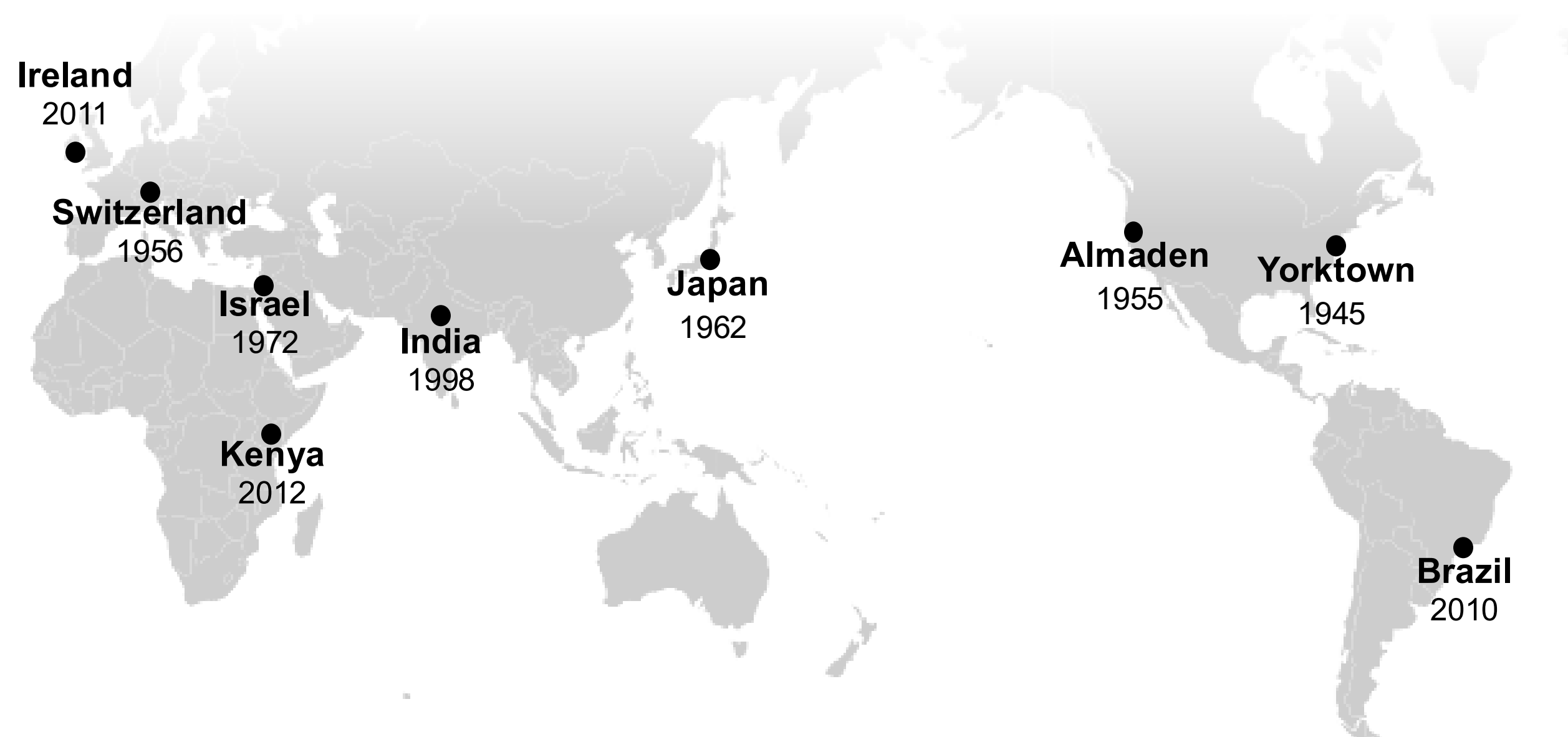
3000 Researchers

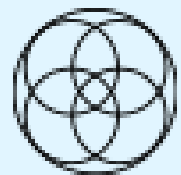
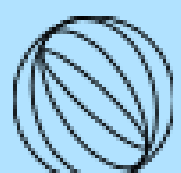
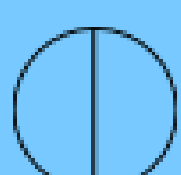
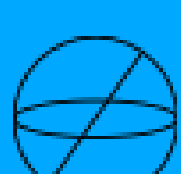
6 Nobel Laureates

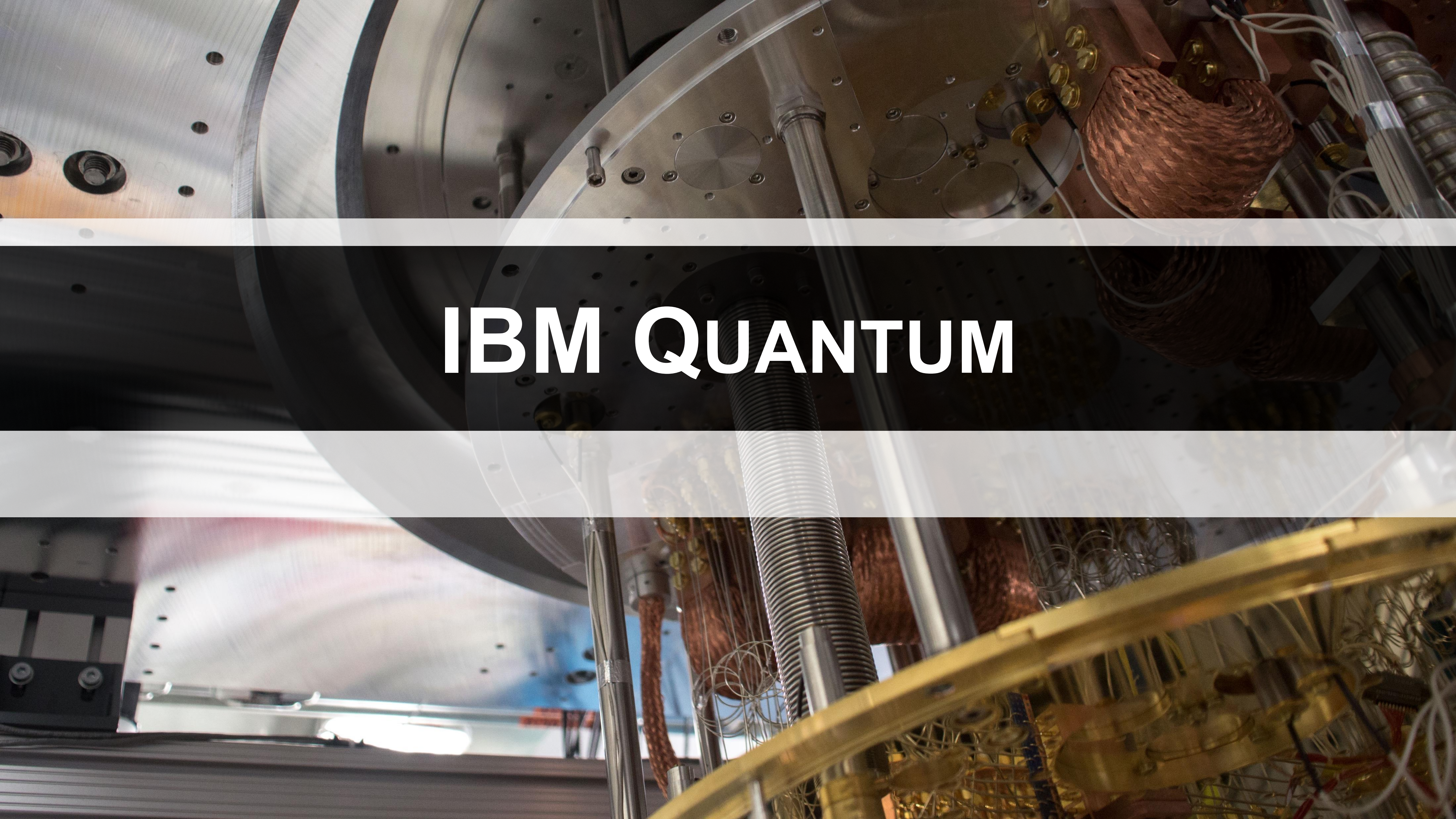
10 US Medals of Technology

5 US Medals of Science

6 Turing Awards

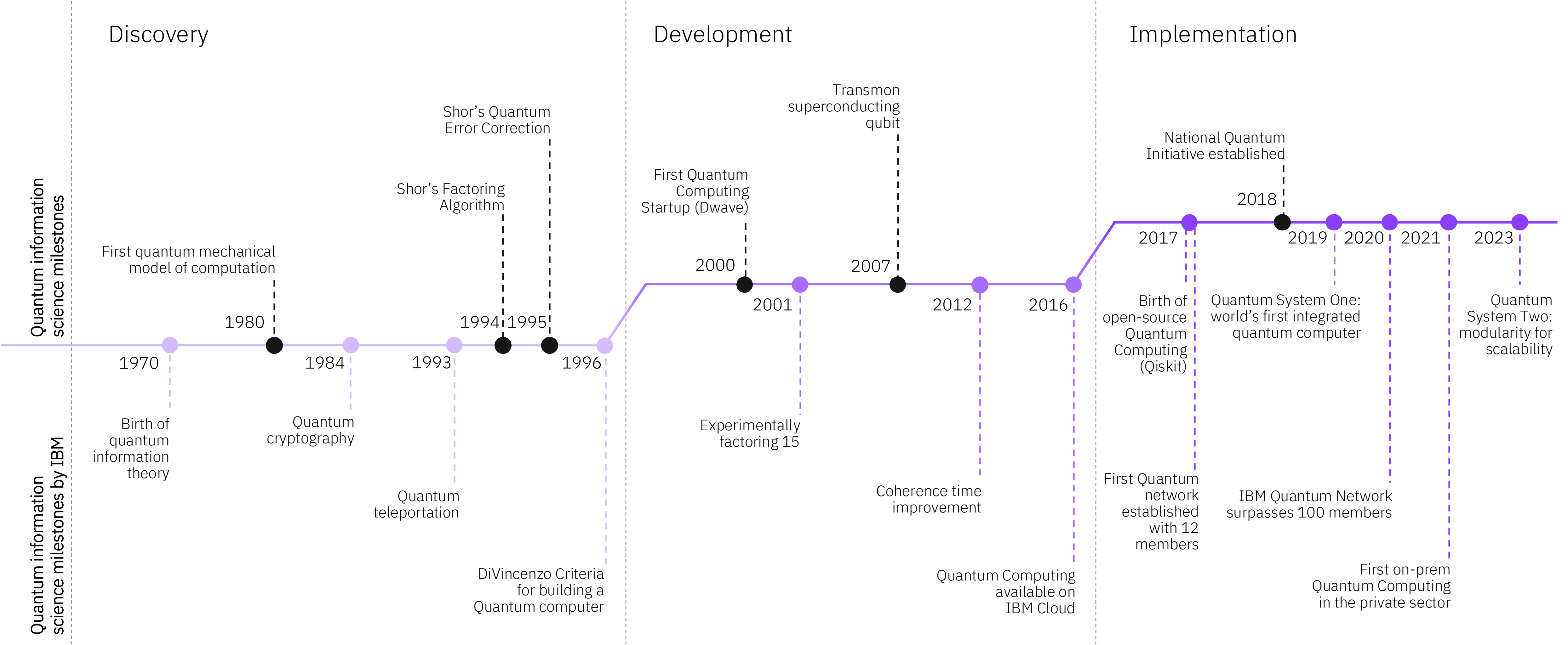


IBM Research		Foundation Models The AI platform
		Hybrid Cloud The software platform for AI & Quantum
		Semiconductor The platform that powers IBM infrastructure
		Quantum The next generation differentiated compute platform

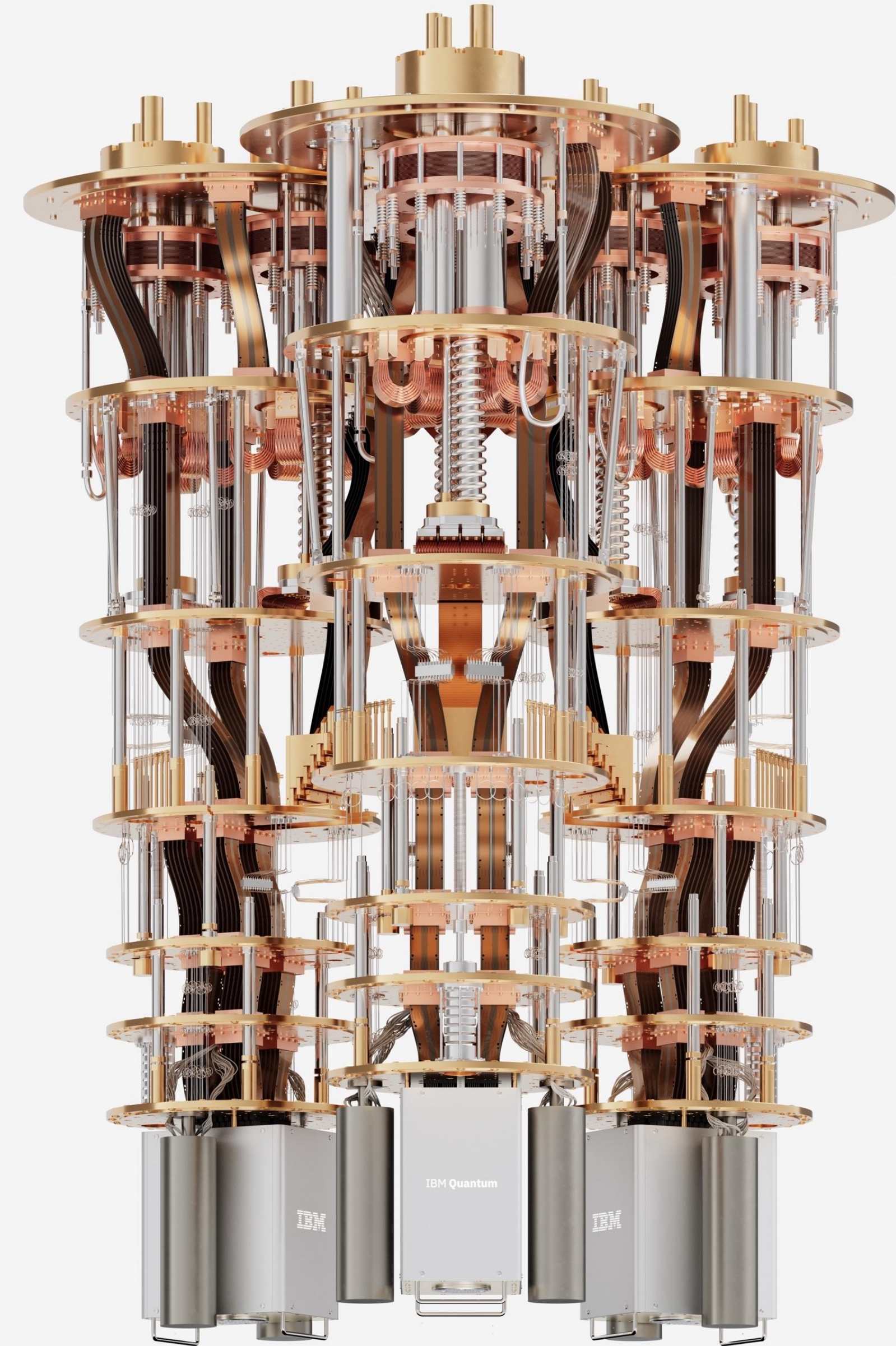


IBM QUANTUM

IBM has been critical to the progression of quantum information science



Bring useful
quantum
computing to
the world



Two key prerequisites for bringing useful quantum computing to the world



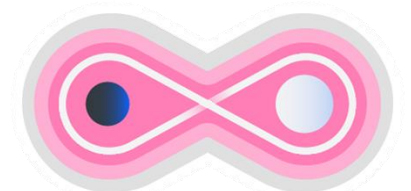
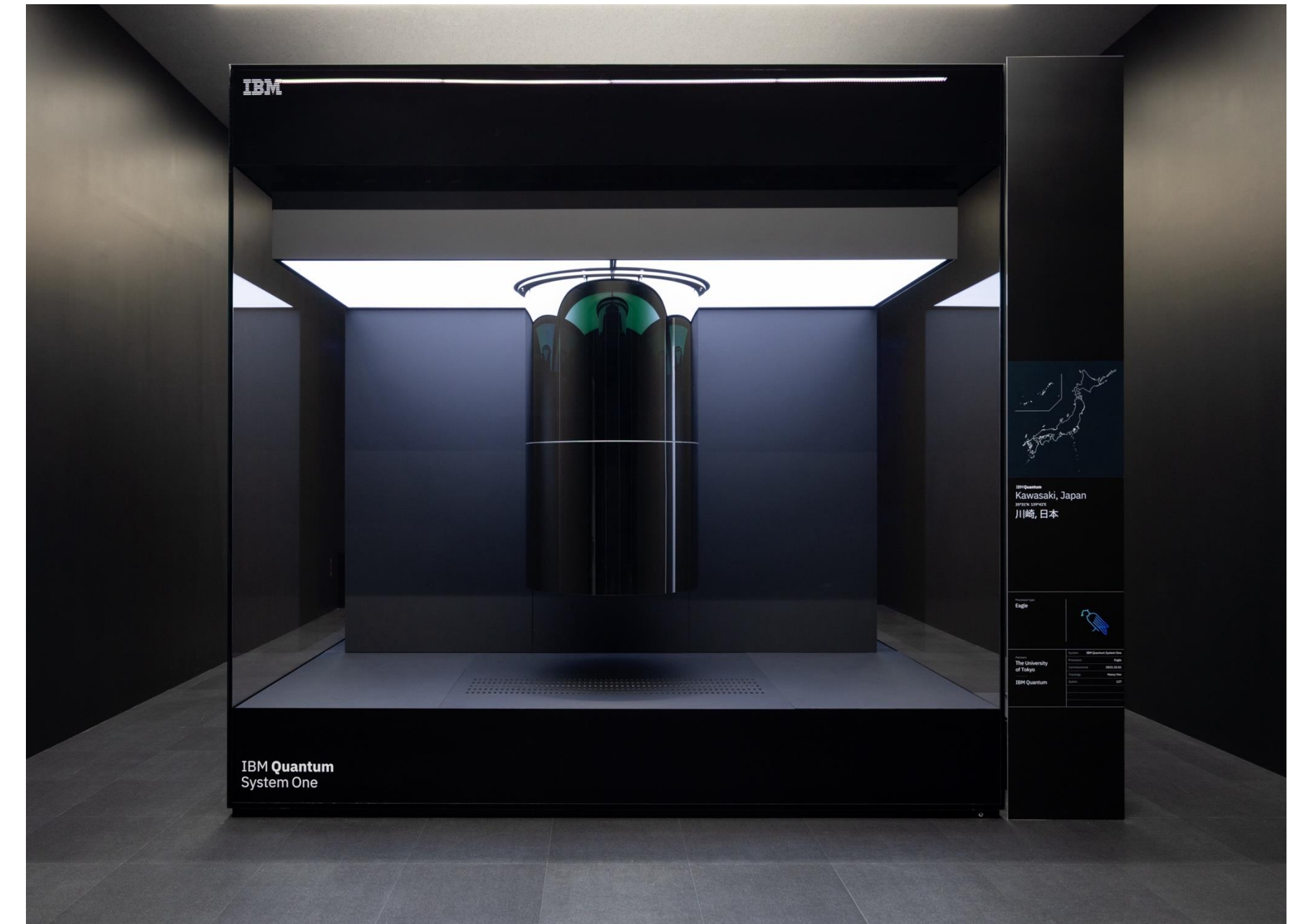
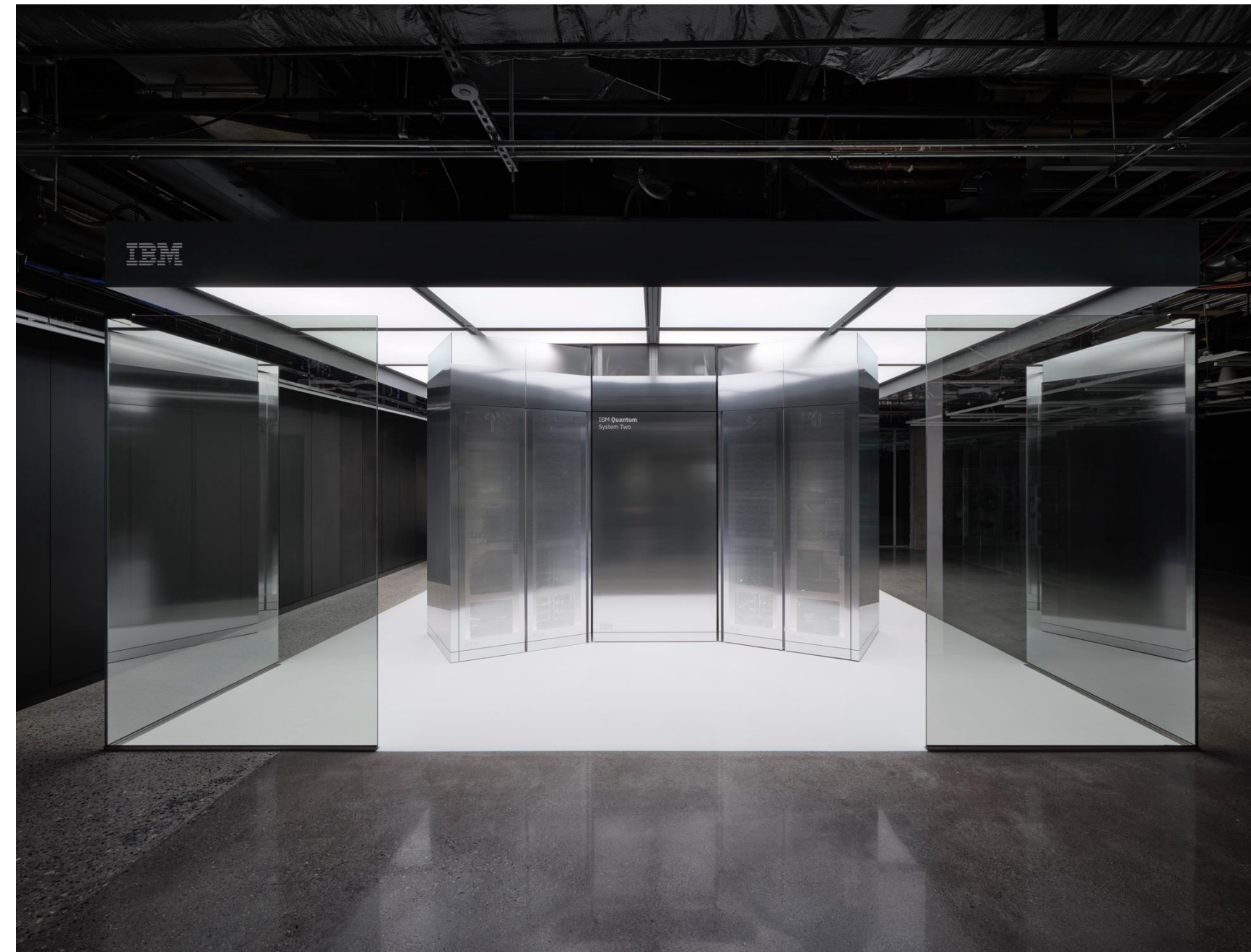
Developing a quantum platform that scales beyond classical computation



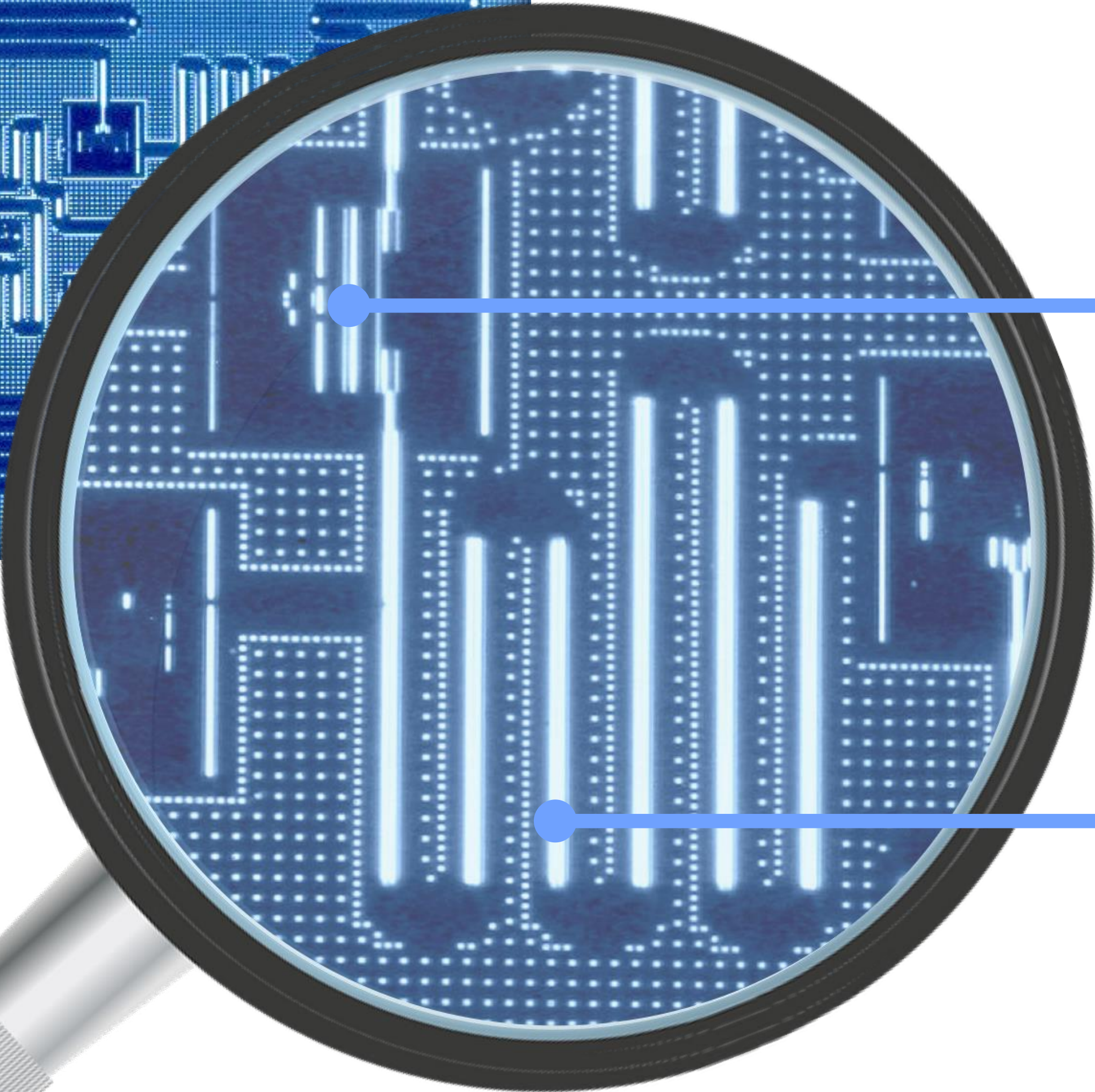
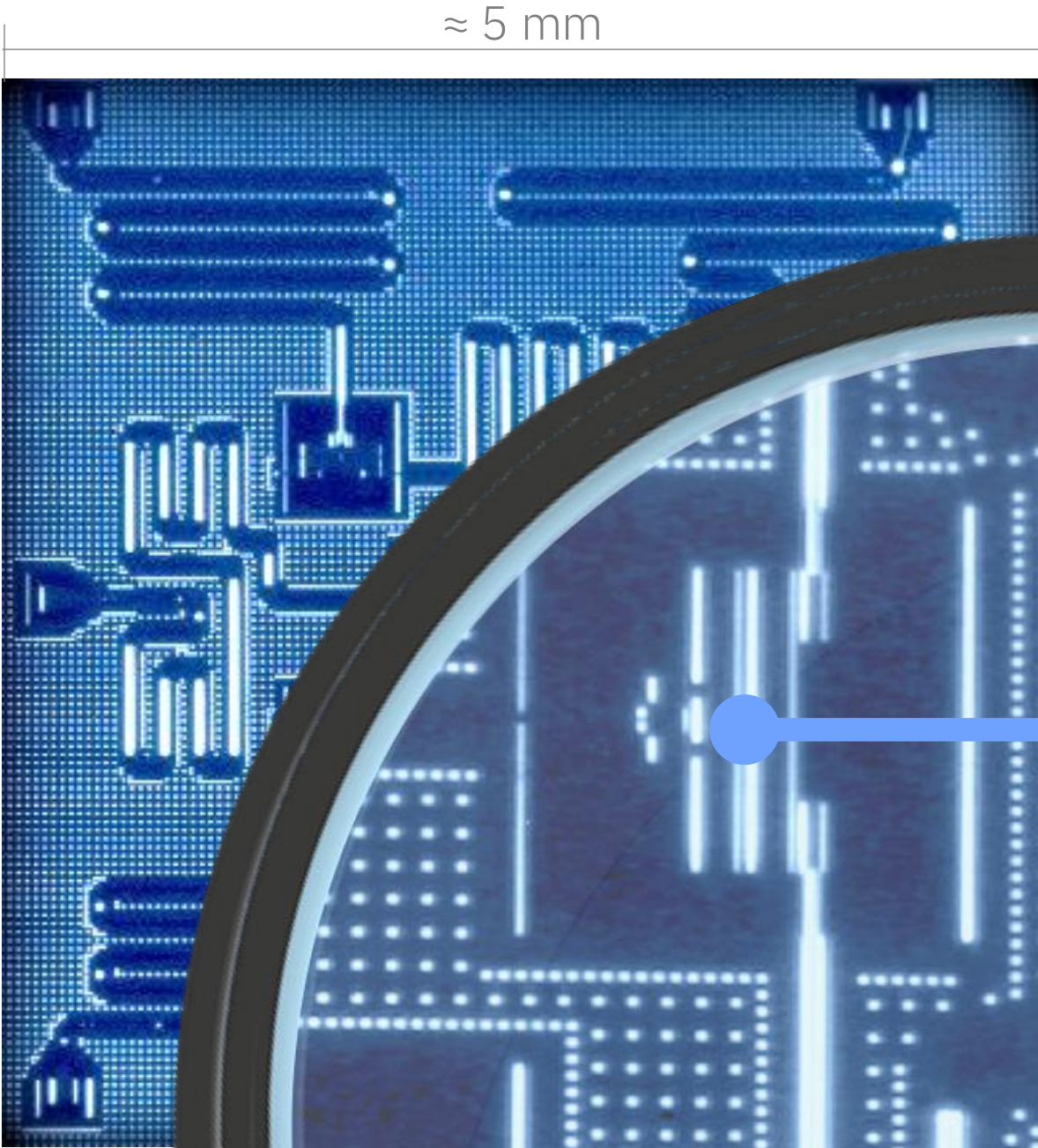
Activating a network of enterprises and academic institutions to advance algorithm discovery



IBM Quantum Platform provides continuous access to **15** 100+ qubit quantum computers to clients across the world

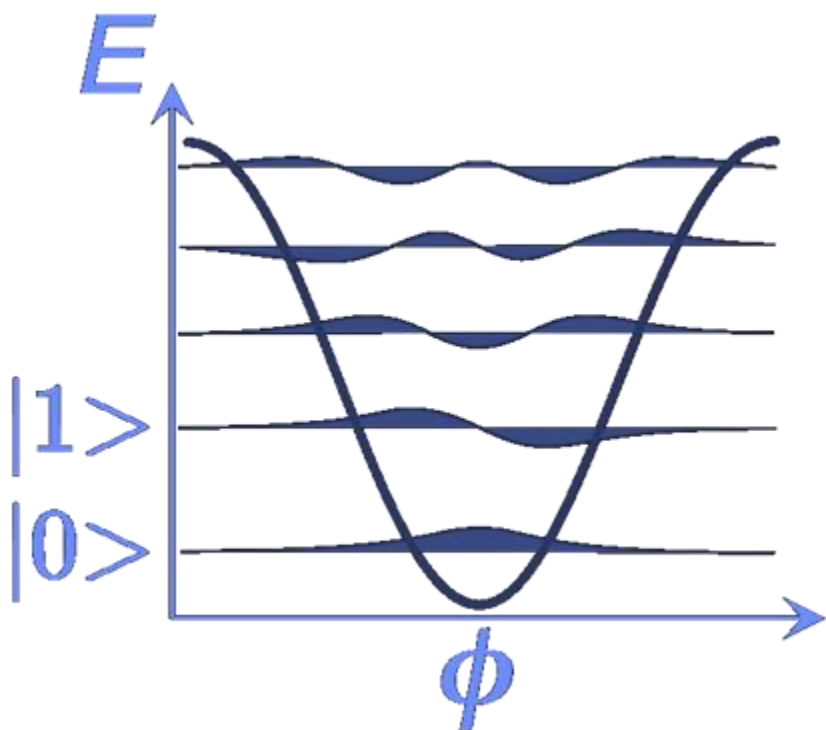
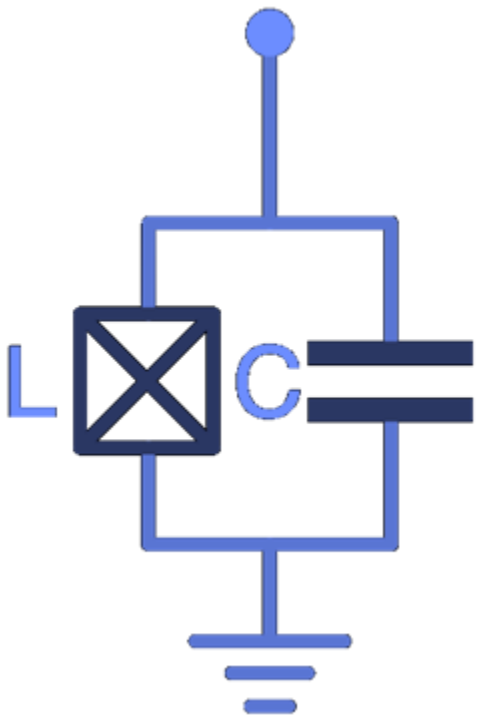
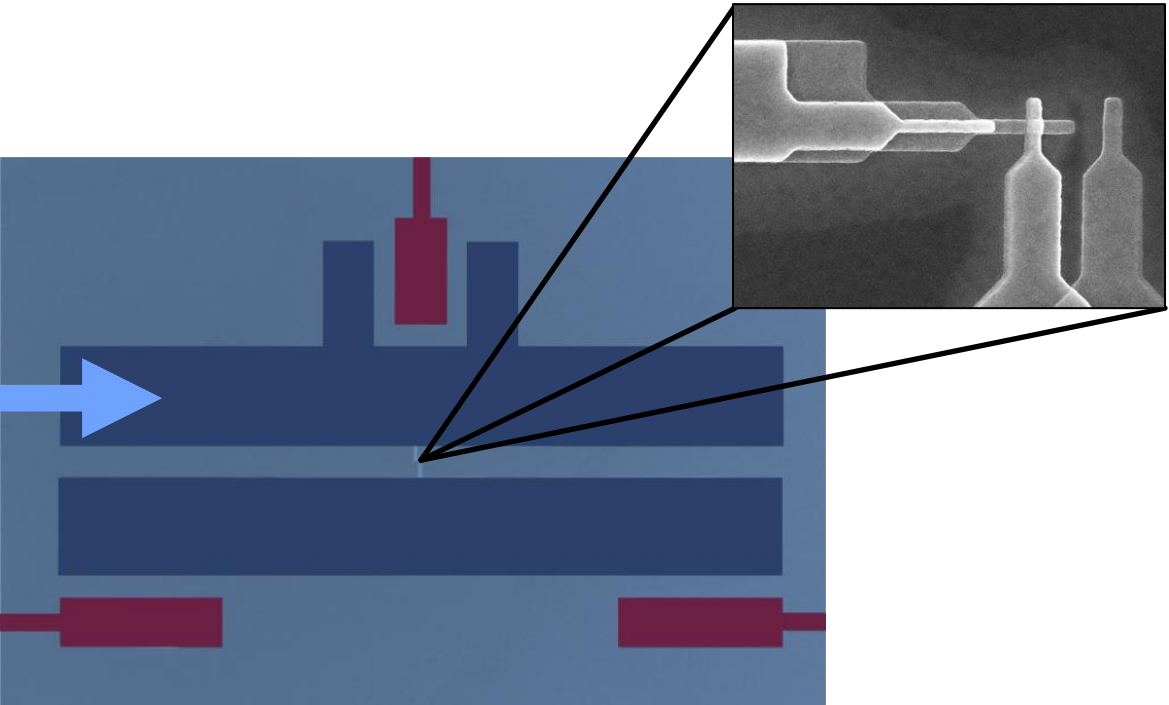


IBM Quantum Processors



Superconducting transmon qubit

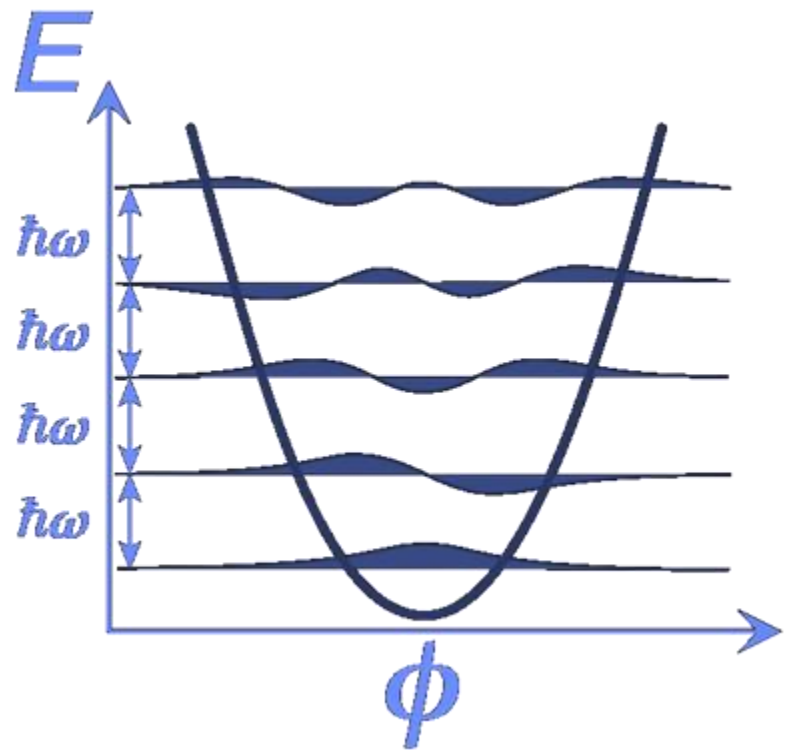
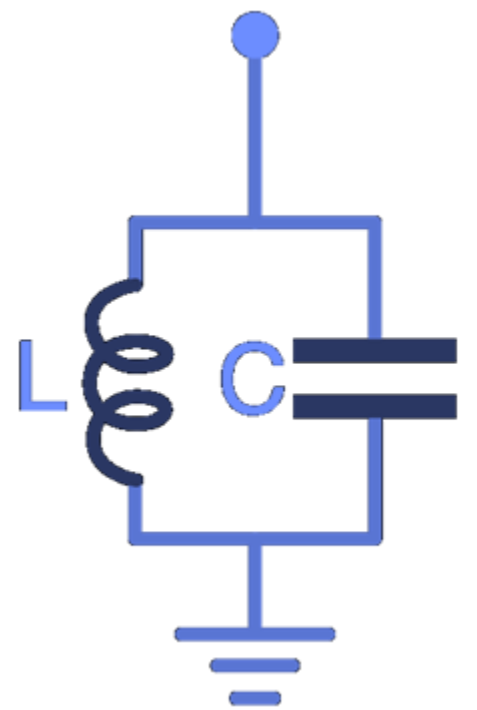
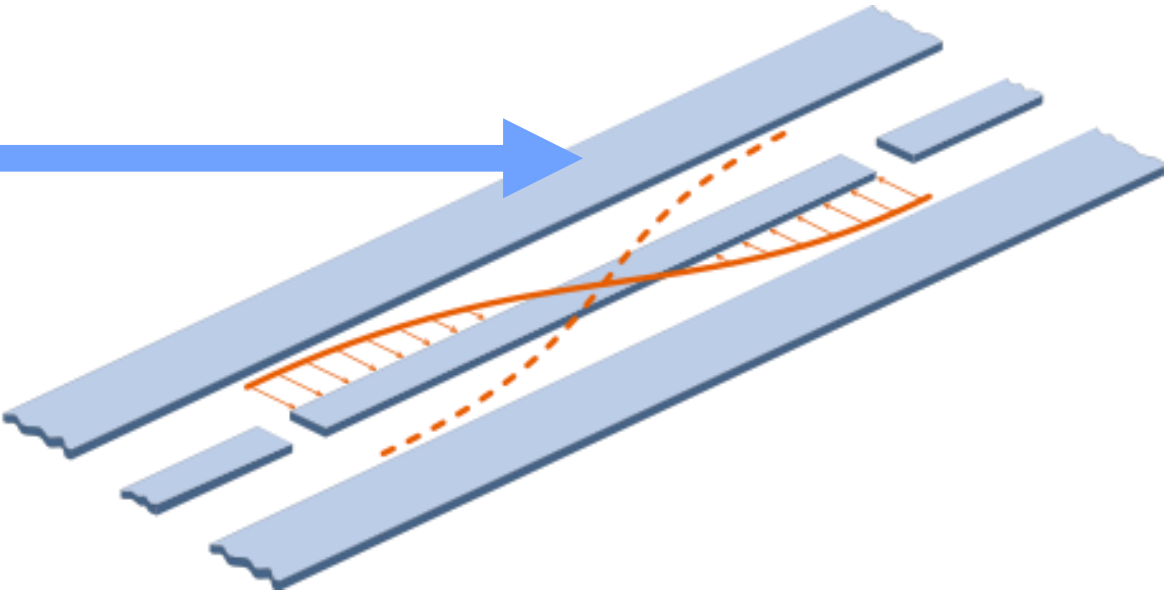
- non-linear Josephson Junction
- anharmonic energy spectrum



$E_{01} \approx 5 \text{ GHz} \approx 240 \text{ mK}$

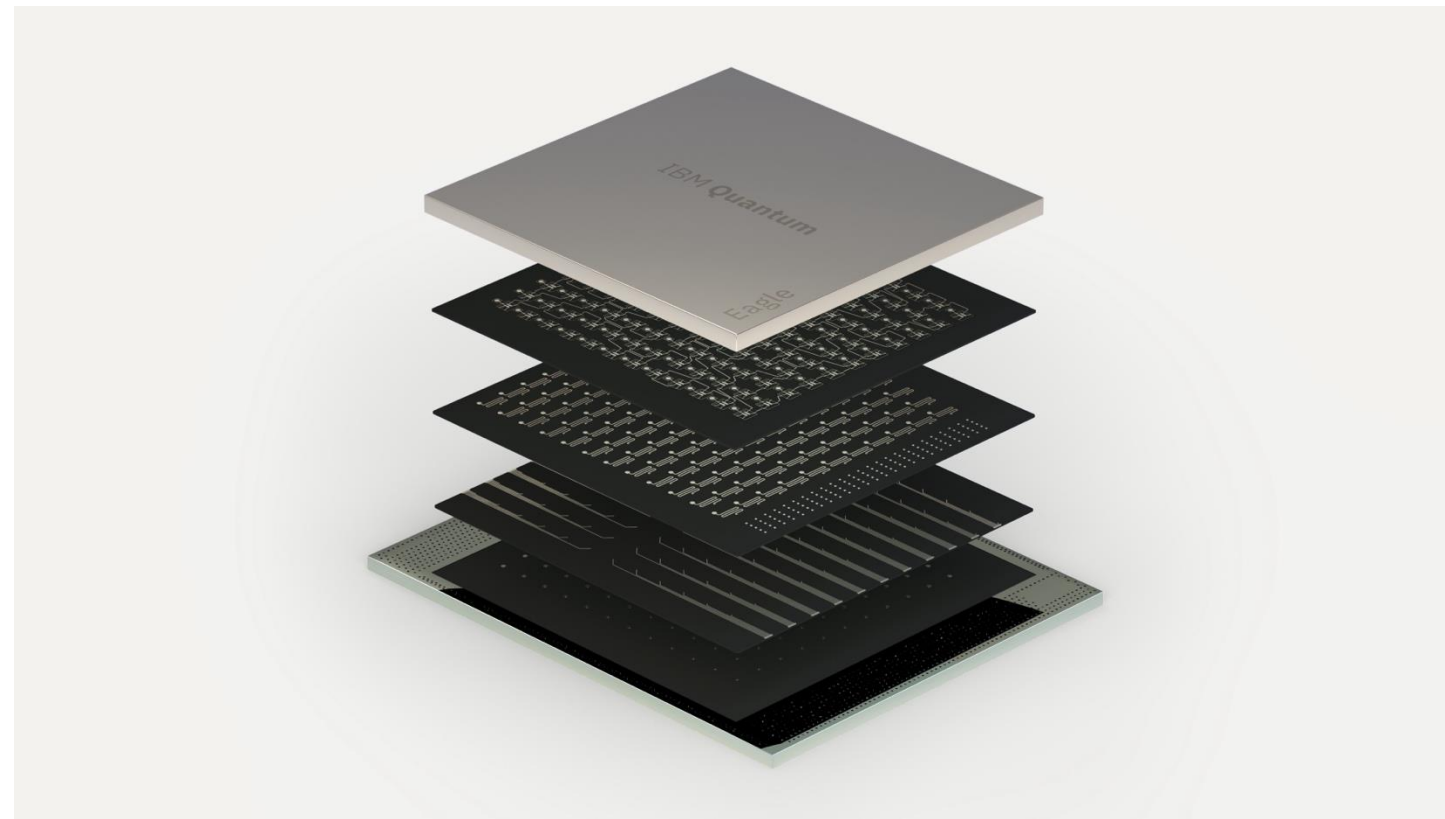
Microwave resonator

- readout of quantum states
- multi-qubit quantum bus



Eagle 2021

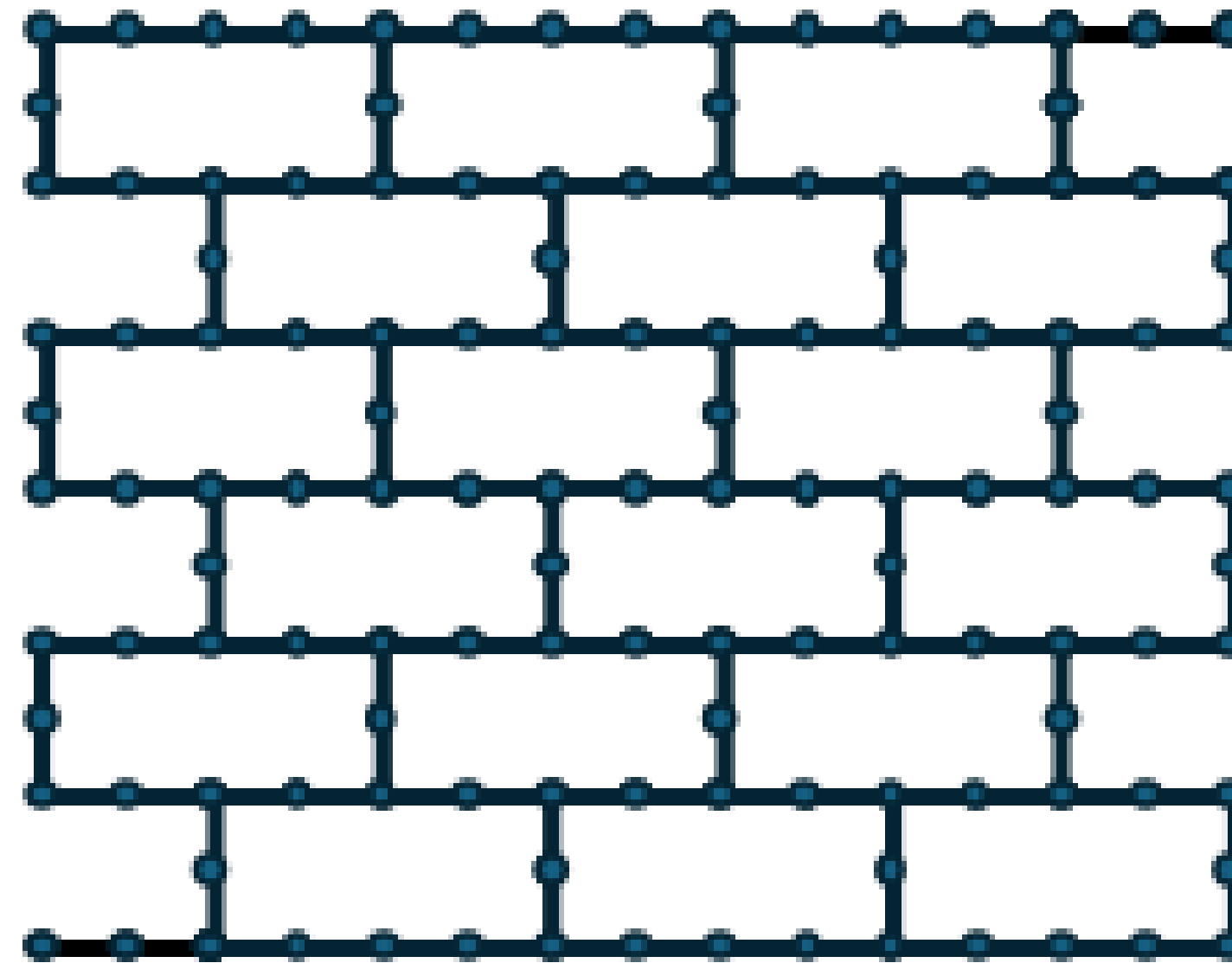
breaking the 100-qubit barrier



127 qubits
fixed frequency couplers
heavy hex topology

Heron 2023

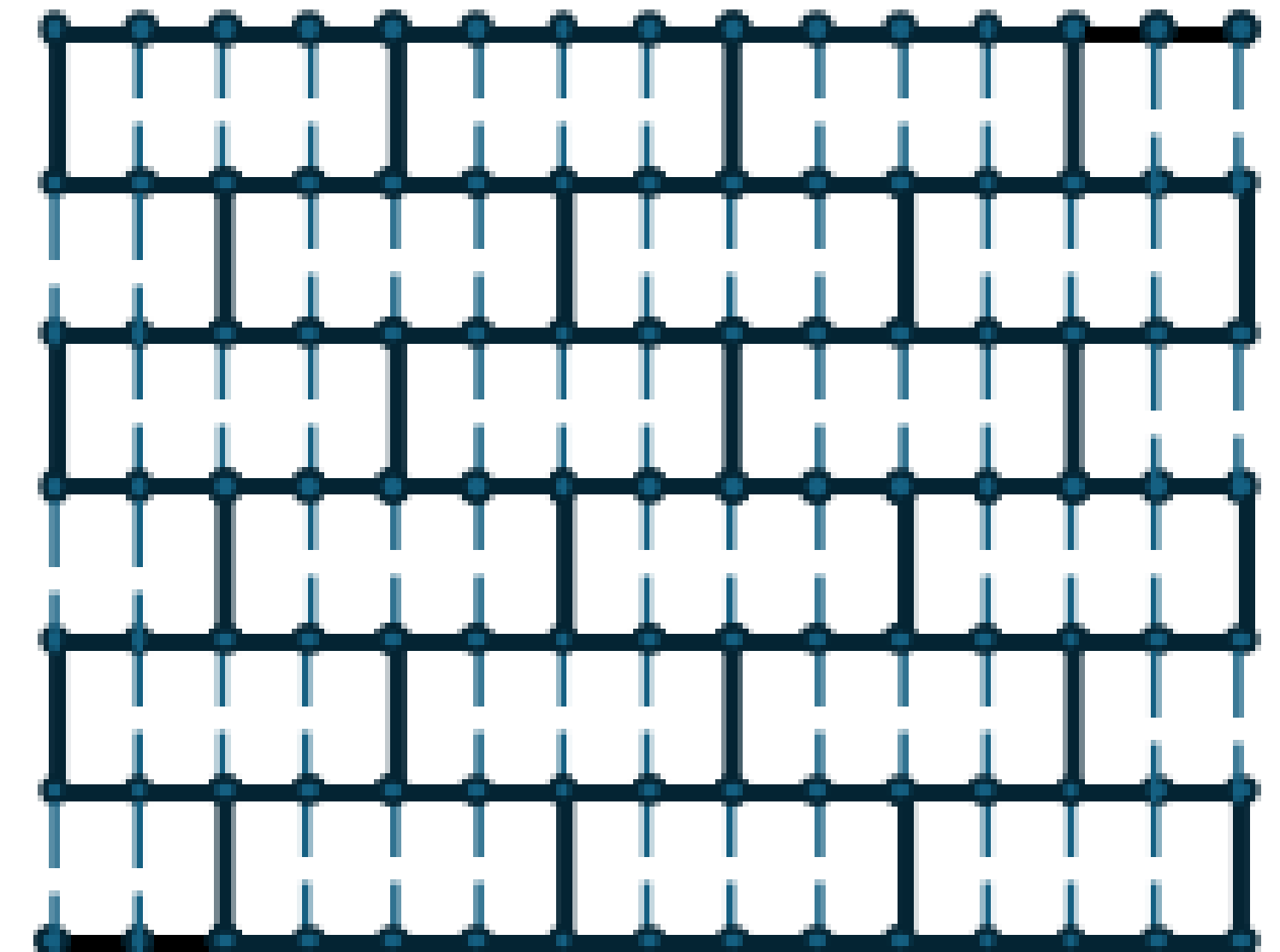
designed for quality



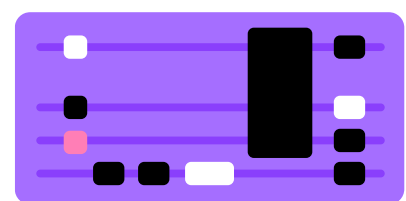
133 / 156 qubits
tunable coupler architecture
heavy hex topology

Nighthawk 2025

higher circuit complexity



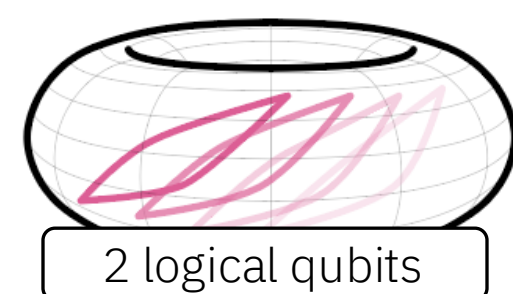
120 qubits
tunable coupler architecture
square lattice topology



IBM fault-tolerant quantum computing roadmap



Loon (2025)

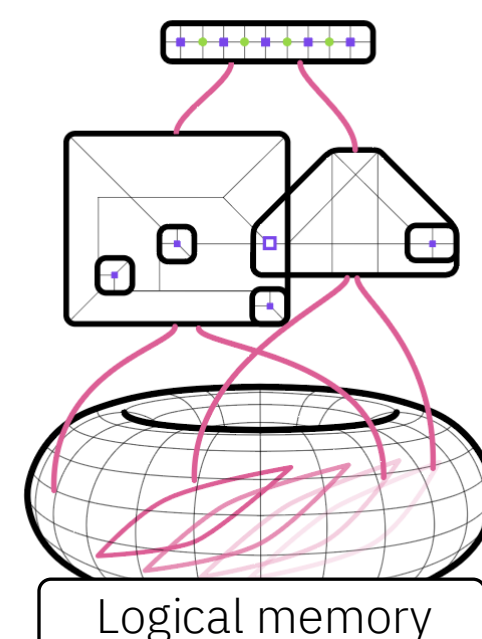


6-way connectivity

c-coupler demo

Automated design

Kookaburra (2026)

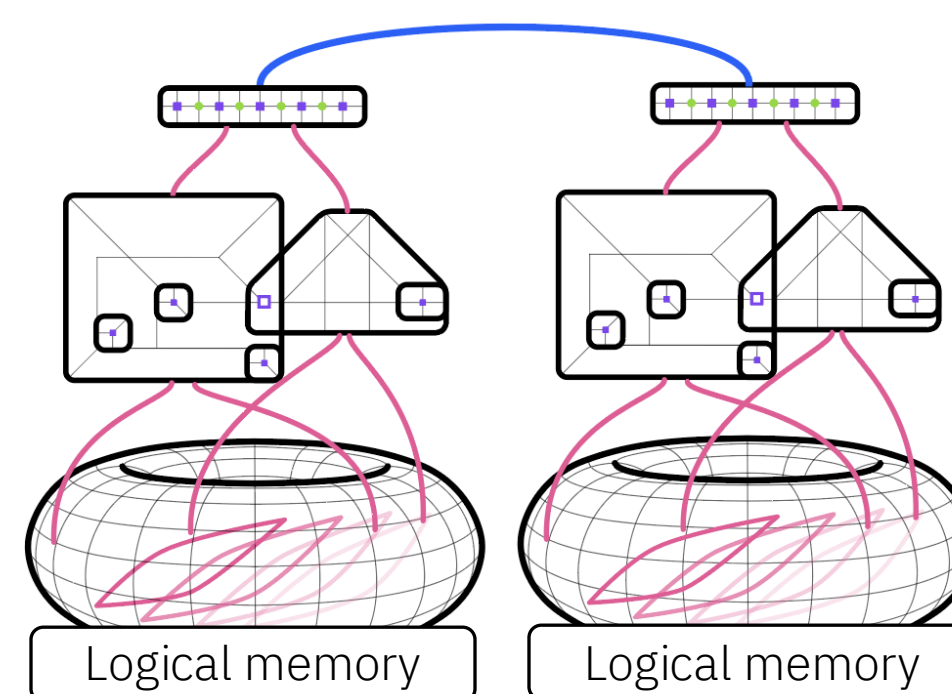


“Long” c-couplers

Real-time decoding

LPU (+~100 qubits) for logical operations

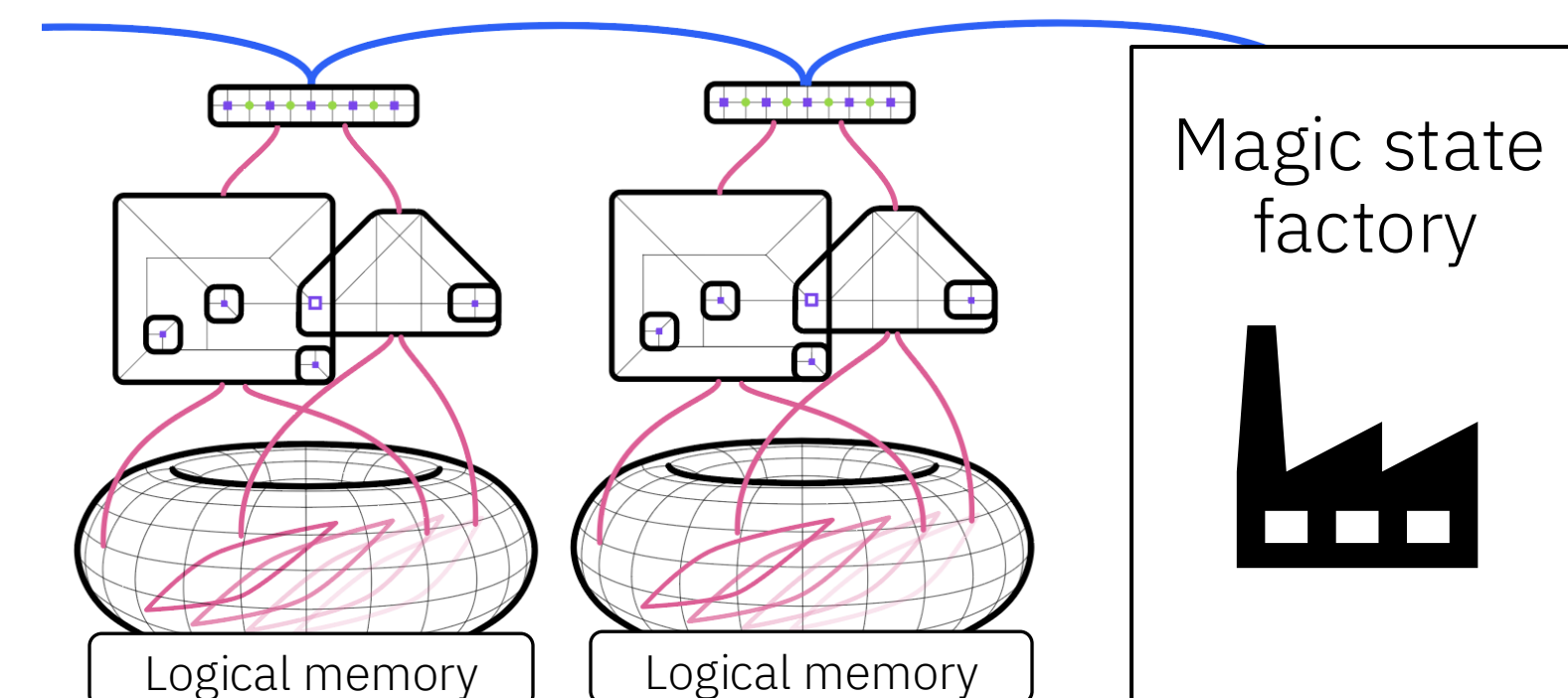
Cockatoo (2027)



Two blocks of gross code + LPU

Module-to-module logical communication over l-couplers

Starling (2028)



Multiple blocks of gross code + LPUs

Universal computation with magic state distillation

IBM Quantum Starling
200 logical qubits
100 million quantum gates
—
2029



IBM Quantum Data Center
Poughkeepsie, New York

IBM Quantum Blue Jay
2000 logical qubits
1 billion quantum gates
—
2033+

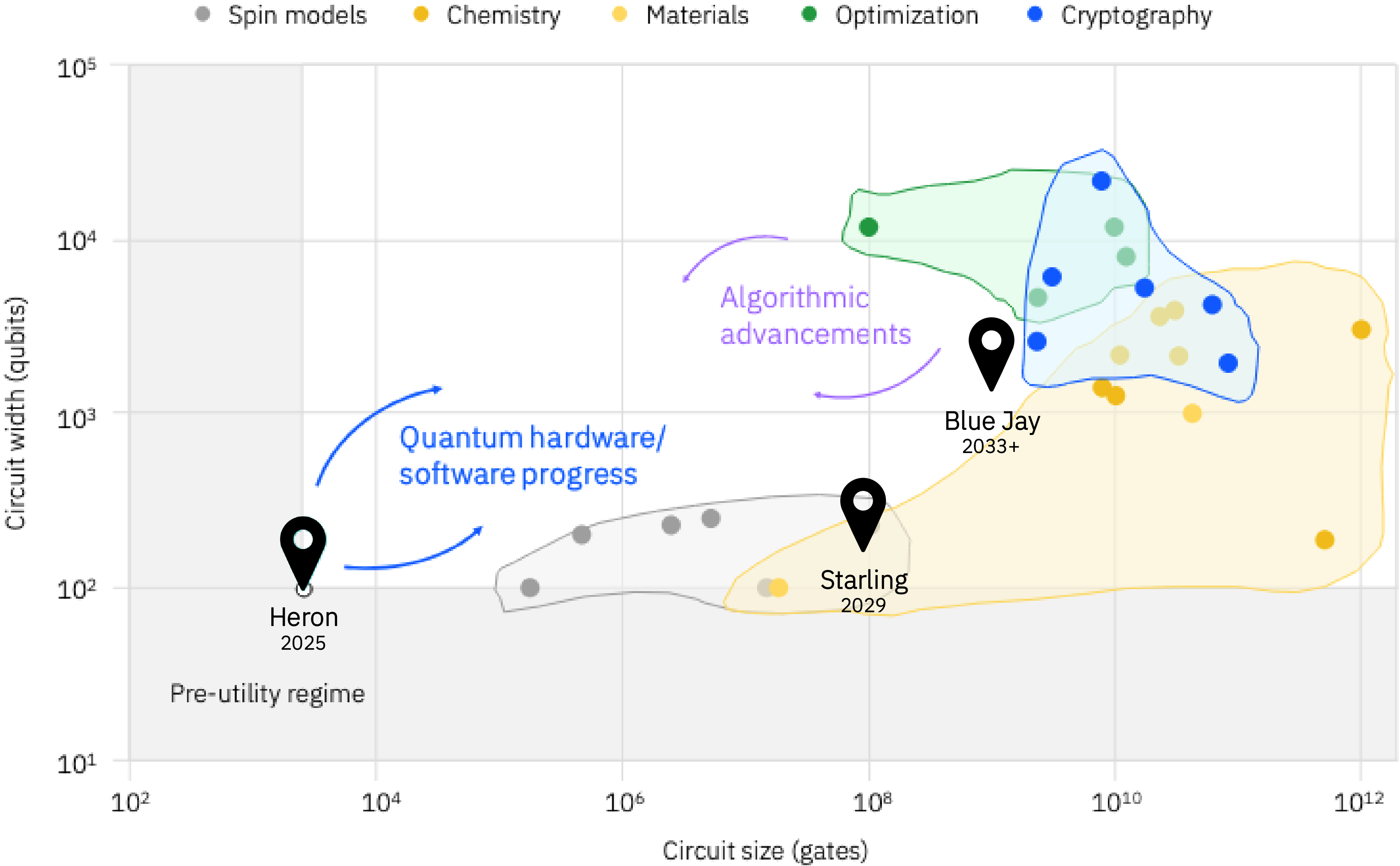
IBM Quantum System Two (4x)
Supports 1000+ physical qubits
15000+ quantum gates
—
2025



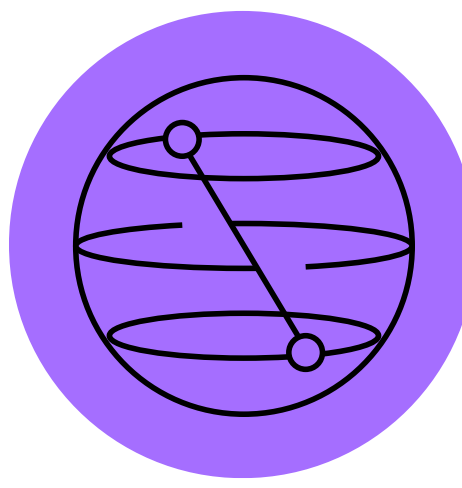
What can you do with a large-scale, fault-tolerant quantum computer?

Investments in **algorithm research** should enable known applications to be realised sooner.

Advances in quantum hardware and software enable new applications or use cases to be discovered, potentially with current or near-future systems



Two key prerequisites for bringing useful quantum computing to the world



Developing a quantum platform that scales beyond classical computation



Activating a network of enterprises and academic institutions to advance algorithm discovery



IBM Quantum Network

A global community driving innovation

275+

Member organizations

40+

Quantum Innovation Centers,
8 with Dedicated Service

125+

Members of QICs
Includes research centers and universities

50+

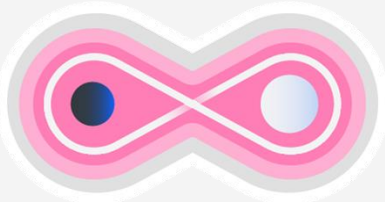
Industry clients
Includes Quantum Accelerator, Premium
Plan, and industry QIC members

50+

Startups

10+

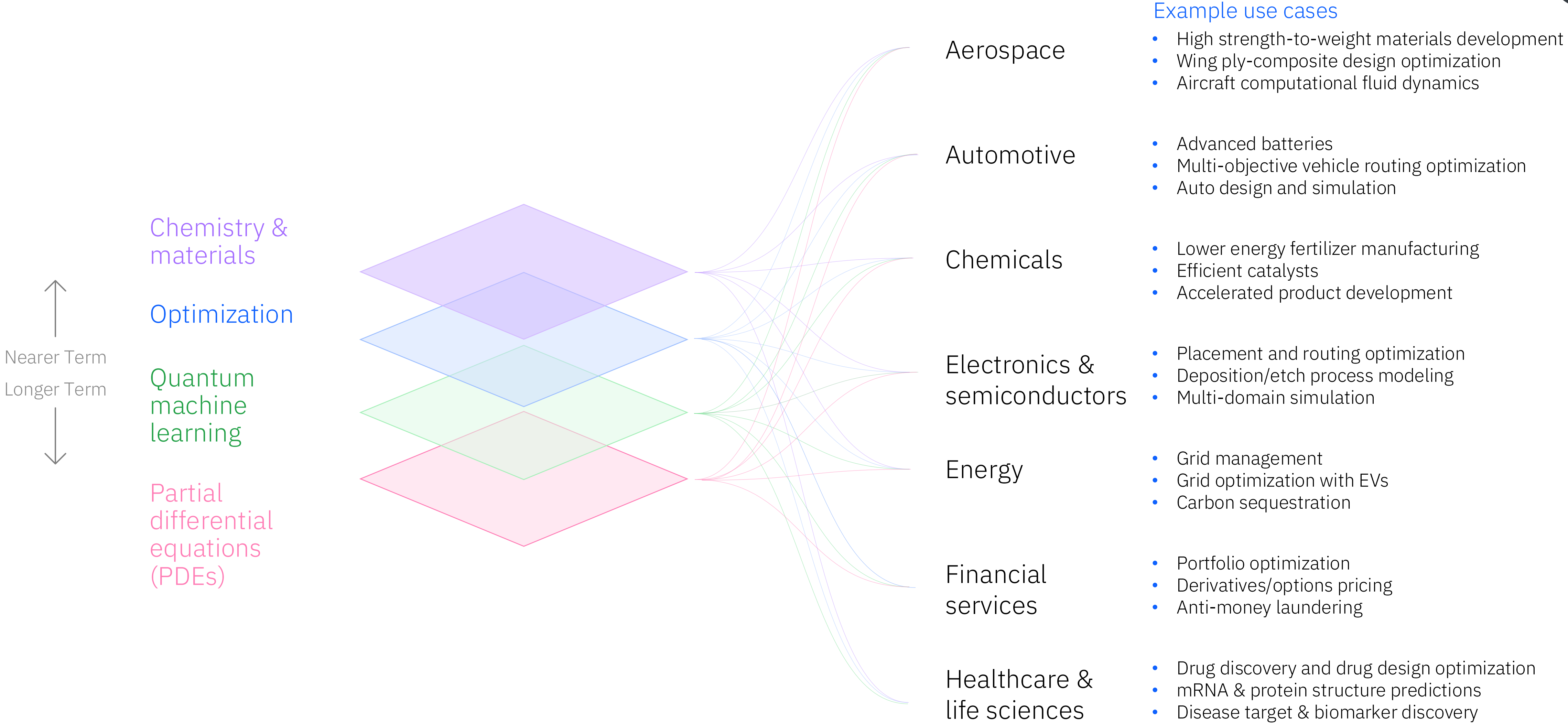
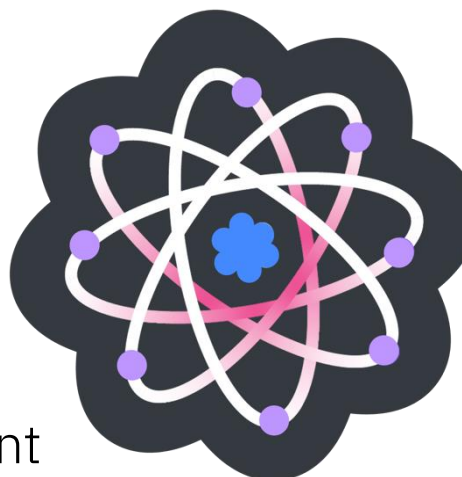
Ecosystem Partners
Includes re-sellers, GSIs, and ISVs



Qiskit Fall Fest 2025

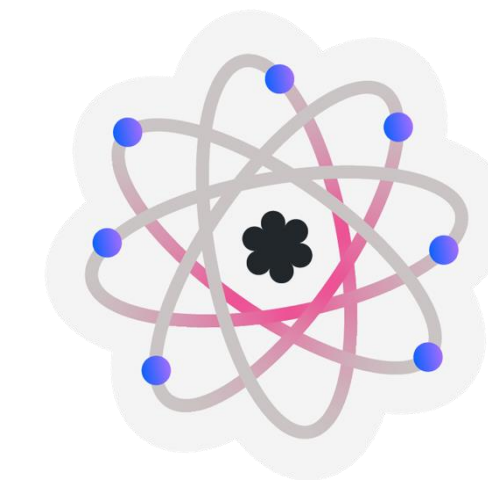


Quantum computing is expected to have impact across industries



Examples of
these problems

Modeling molecules, atoms, electrons,
and quarks with **unprecedented accuracy**



Developing lighter, longer-lasting batteries for electric vehicles, electronics, and energy grid storage



Designing lighter, stronger materials to allow planes to be more efficient and to need less maintenance



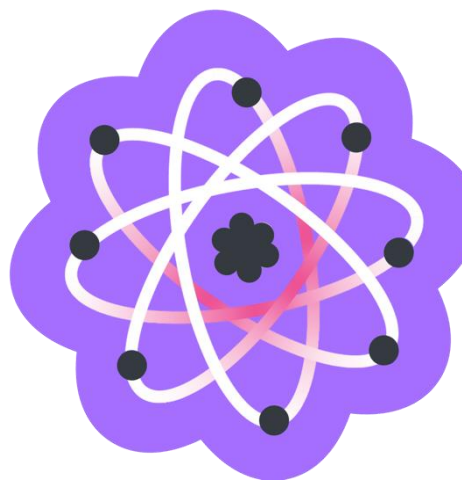
Discovering new classes of antibiotics to counter the emergence of multidrug-resistant bacterial strains



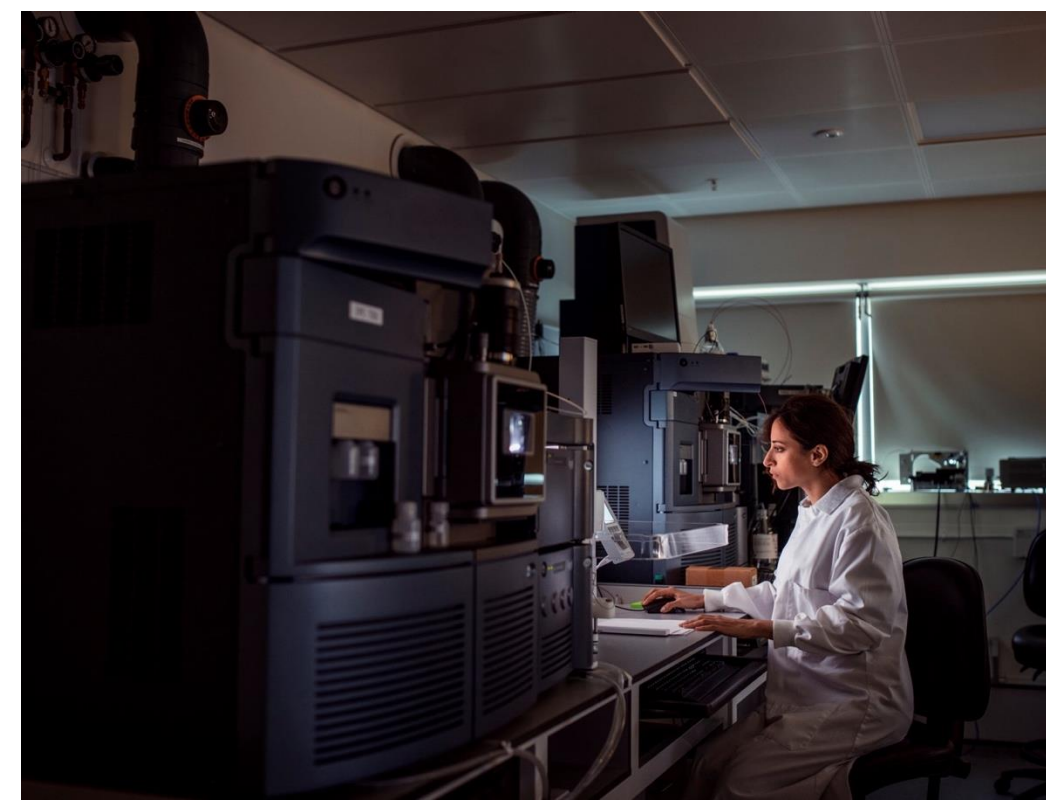
Designing optimal superconductors for MRI, electromobility, and renewable energies

Examples of
these problems

Solving algebra in exponential spaces.
Finding **hidden patterns** in structured problems.



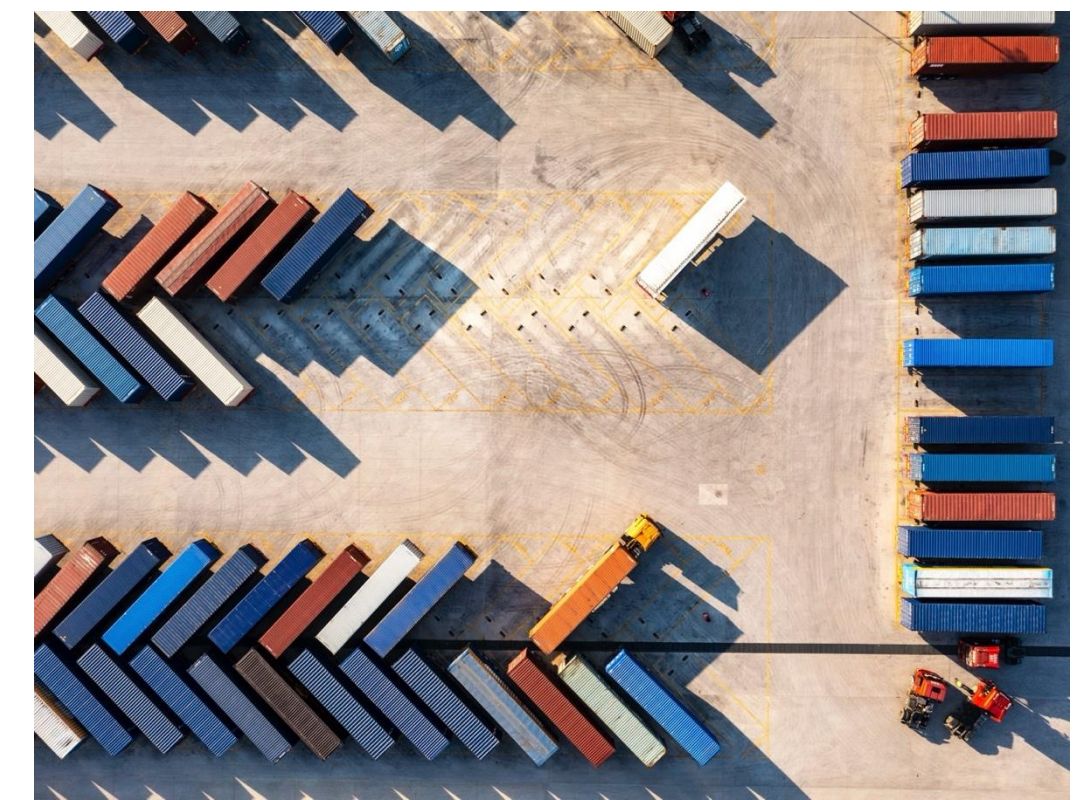
Improving understanding
of physical and chemical
properties of materials



Improving patient
outcomes by designing
optimal cell-centric
therapeutics

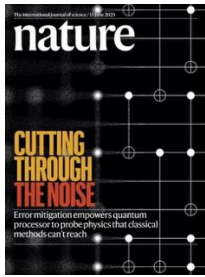


Strengthening risk
management through
better time series and
sequence prediction



Optimizing vehicle routing
and scheduling for large-
scale logistics networks

Algorithms connect computational capabilities to industry applications...

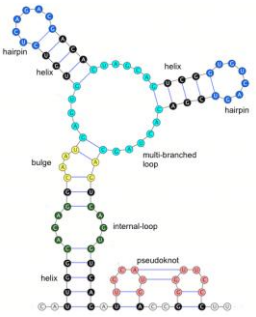


Evidence for the utility of quantum computing before fault tolerance

Nature, 618, 500 (2023)

127 qubits

simulation

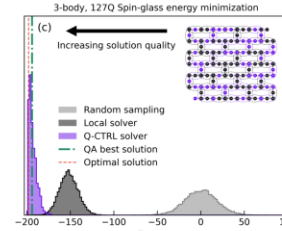


mRNA secondary structure prediction using utility-scale quantum computers

arXiv:2405.20328

80 qubits

simulation

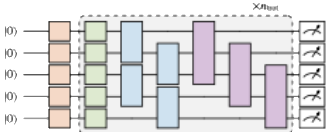


Quantum optimization using a 127-qubit gate-model IBM quantum computer can outperform quantum annealers for nontrivial binary optimization problems

arXiv:2406.01743

127 qubits

optimization

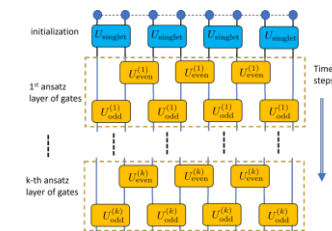


Bias-Field Digitized Counterdiabatic Quantum Algorithm for Higher-Order Binary Optimization

arXiv:2409.04477

156 qubits

optimization

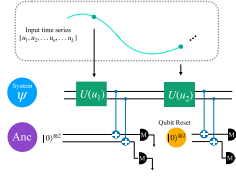


Simulating large-size quantum spin chains on cloud-based superconducting quantum computers

Phys. Rev. Research 5, 013183 (2023)

102 qubits

simulation

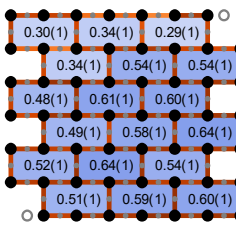


Quantum reservoir computing with repeated measurements on superconducting devices

arXiv:2310.06706

120 qubits

QML

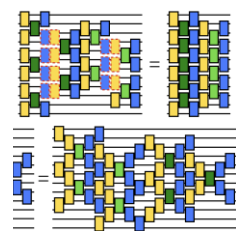


Realizing the Nishimori transition across the error threshold for constant-depth quantum circuits

Nature Physics, 21, 161-167 (2025)

125 qubits

simulation

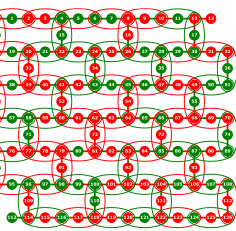


Scalable Circuits for Preparing Ground States on Digital Quantum Computers: The Schwinger Model Vacuum on 100 Qubits

PRX Quantum 5, 020315 (2024)

100 qubits

simulation

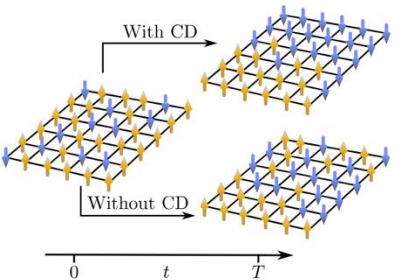


Scaling Whole-Chip QAOA for Higher-Order Ising Spin Glass Models on Heavy-Hex Graphs

arXiv:2312.00997

127 qubits

optimization

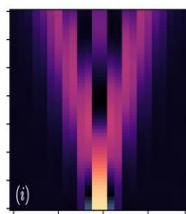


Digitized counterdiabatic quantum critical dynamics

arXiv:2502.15100

156 qubits

optimization

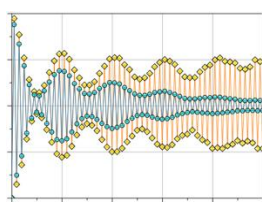


Quantum Simulations of Hadron Dynamics in the Schwinger Model using 112 Qubits

Phys. Rev. D 109, 114510 (2024)

112 qubits

simulation

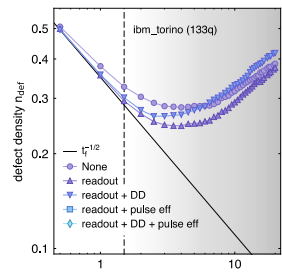


Unveiling clean two-dimensional discrete time quasicrystals on a digital quantum computer

arXiv:2403.16718

133 qubits

simulation



Benchmarking digital quantum simulations and optimization above hundreds of qubits using quantum critical dynamics

arXiv:2404.08053

133 qubits

simulation

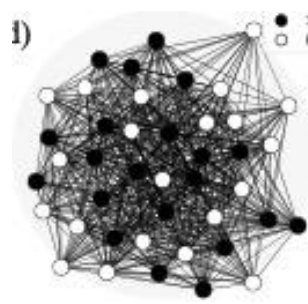


Chemistry beyond the scale of exact diagonalization on a quantum-centric supercomputer

Science Advances 11, 25 (2025)

77 qubits

simulation



Towards a universal QAOA protocol: Evidence of quantum advantage in solving combinatorial optimization problems

arXiv:2405.09169

109 qubits

optimization

Algorithms connect computational capabilities to industry applications



Exploring material degradation caused by radicals

arXiv:2508.08229

Hamiltonian simulation



Simulating supramolecular interactions in pharma-relevant systems

arXiv:2410.09209

Hamiltonian simulation



Predicting mRNA secondary structures with quantum optimization techniques

arXiv:2405.20328

Optimization



Portfolio optimization for fixed income assets

arXiv:2508.13557

Optimization



Optimizing vehicle-to-grid fleet charge scheduling

arXiv:2407.20802
arXiv:2412.07915

Optimization

QML



Simulating advection-diffusion and similar partial differential equations

arXiv:2508.16728

PDEs

IBM Quantum Network in Australia

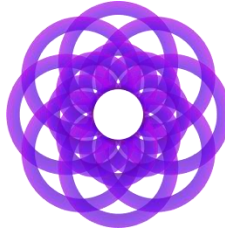


University of Melbourne



IBM Quantum Hub is supported by 3 faculties,
involving 12+ academics,
who have published 50+ research papers,
graduated 14+ Masters and PhD students,
and taught 800+ students in subjects using IBM
Quantum Systems

Q-CTRL



Five quantum computing
world records set on IBM
systems with Q-CTRL
software

33 qubits¹

World's largest QPE demo

40 qubits²

World's longest-range gate
teleportation

75 qubits³

World's largest verifiable
entangled state

104 qubits⁴

World's largest constrained
quantum optimization

156 qubits⁵

World's largest successful
quantum optimization

1) [arXiv:2408.04946](https://arxiv.org/abs/2408.04946)
2) PRX Quantum 6, 020331 (2025)
3) PRX Quantum 6, 020331 (2025)
4) To be published, Network Rail case study
5) arXiv: 2406.01743

IBM Quantum in Australia



IBM Quantum is an industry partner



IBM Quantum is a member



IBM represents AIIA on the QT001 Committee

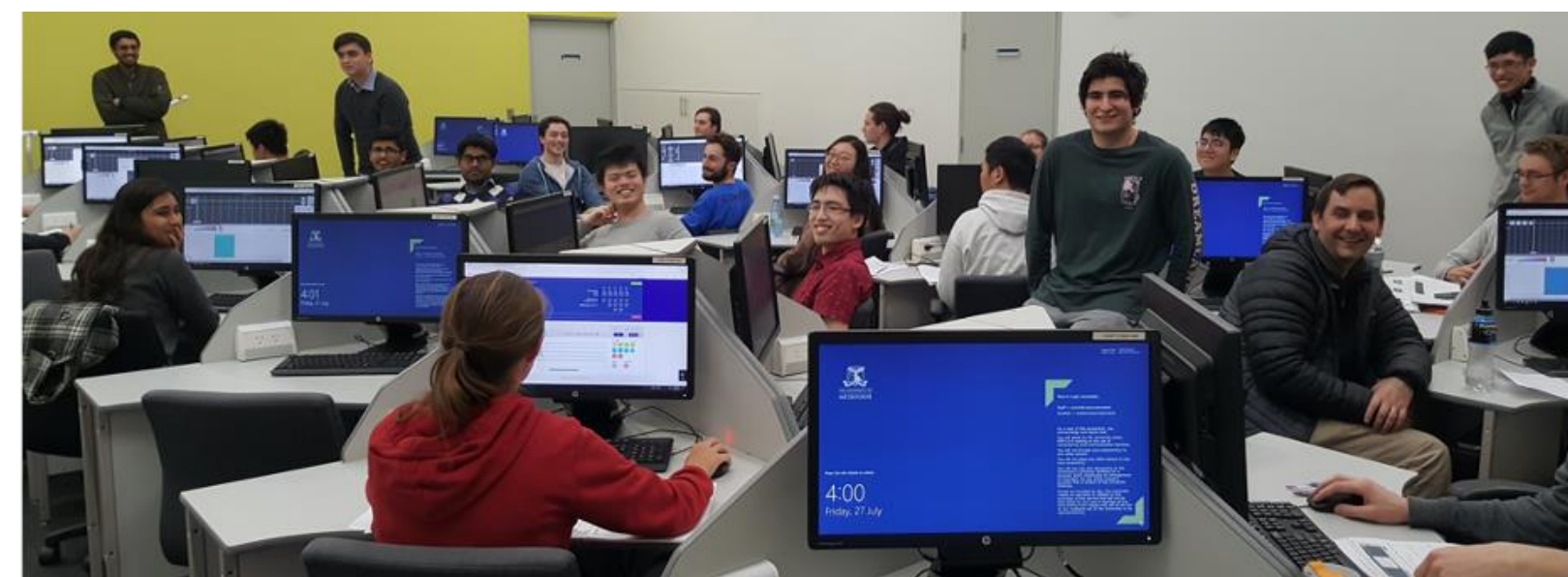


IBM Quantum sponsored '22, '23, '24



IBM Quantum sponsored Summer Camps '22, '23, '24

IBM Quantum used for education and research



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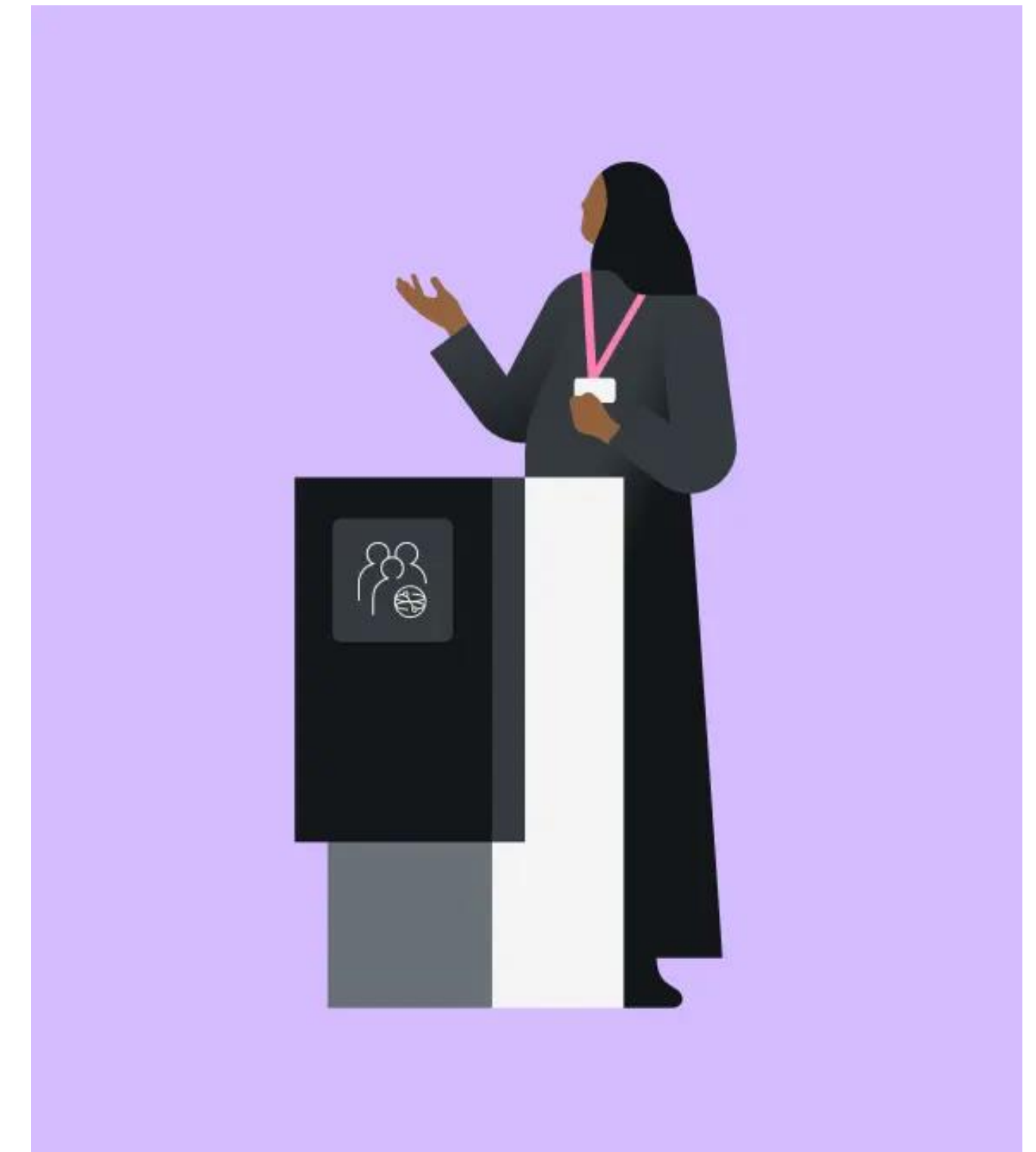
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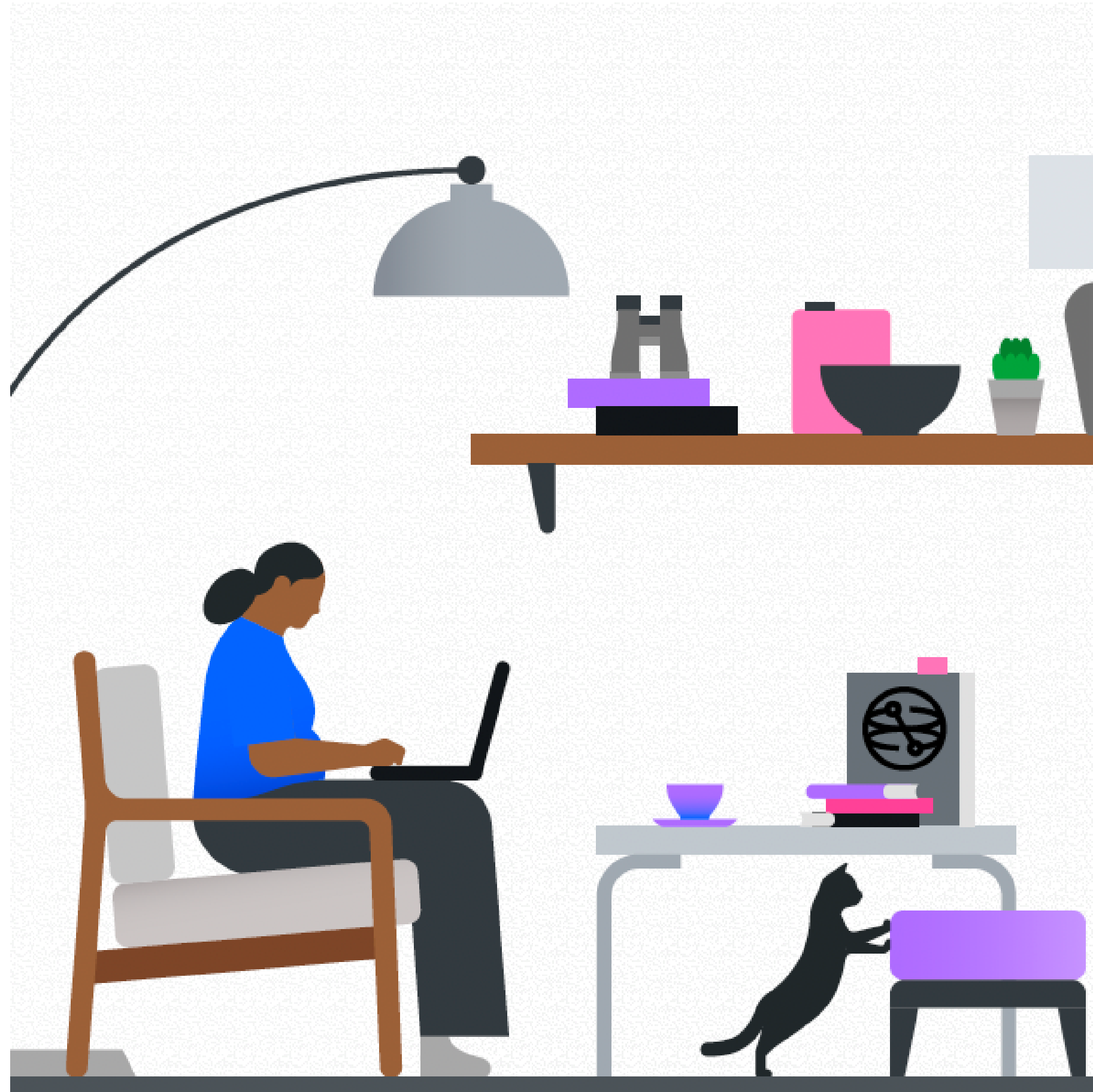
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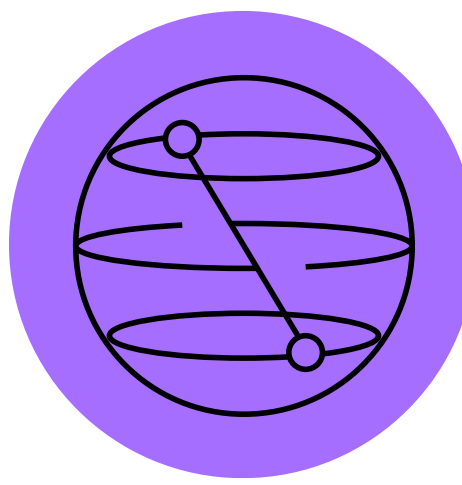


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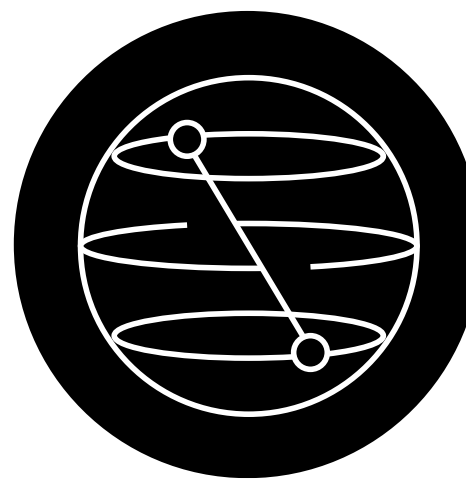
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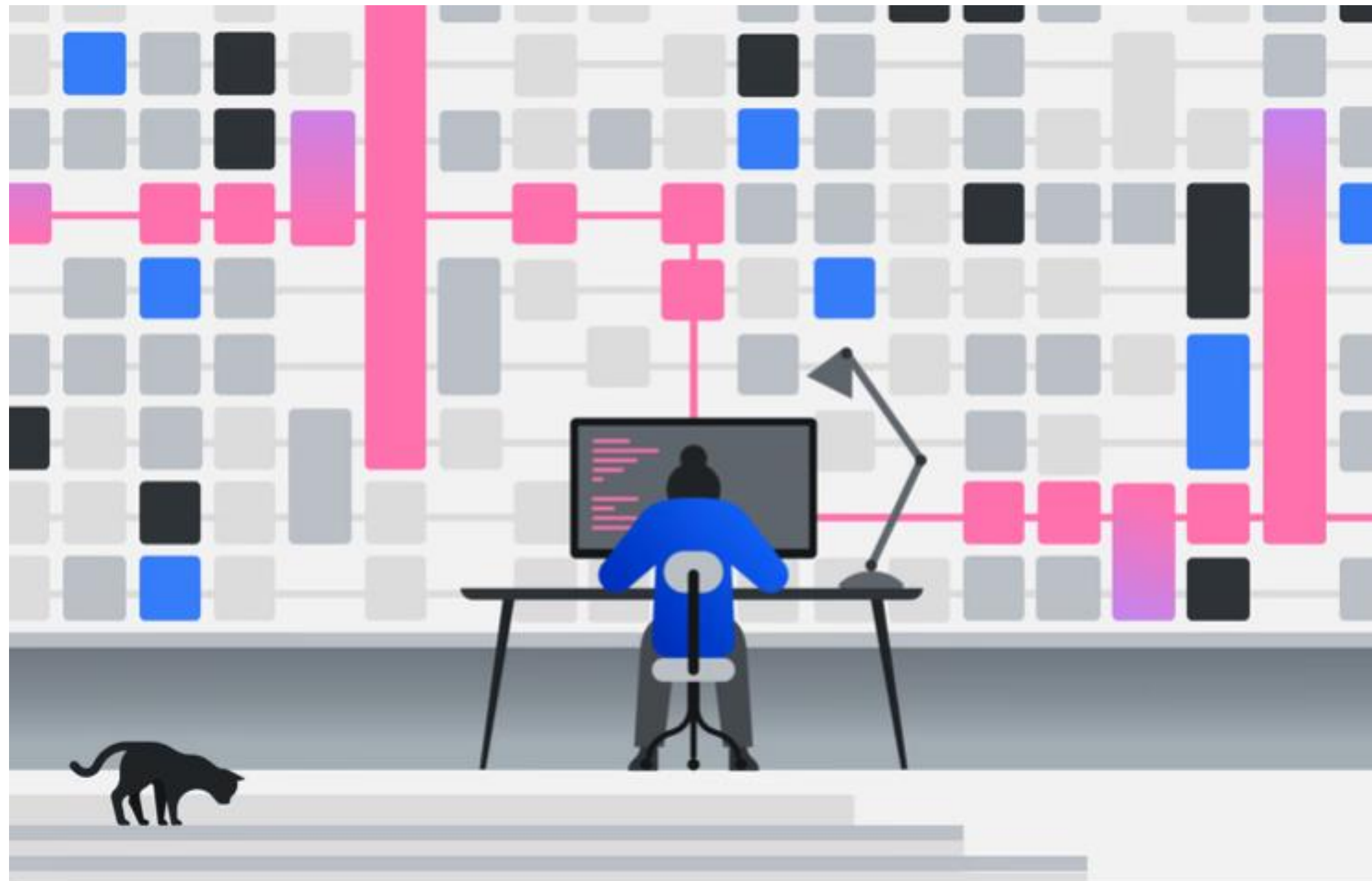
IBM Quantum interns, class of 2023

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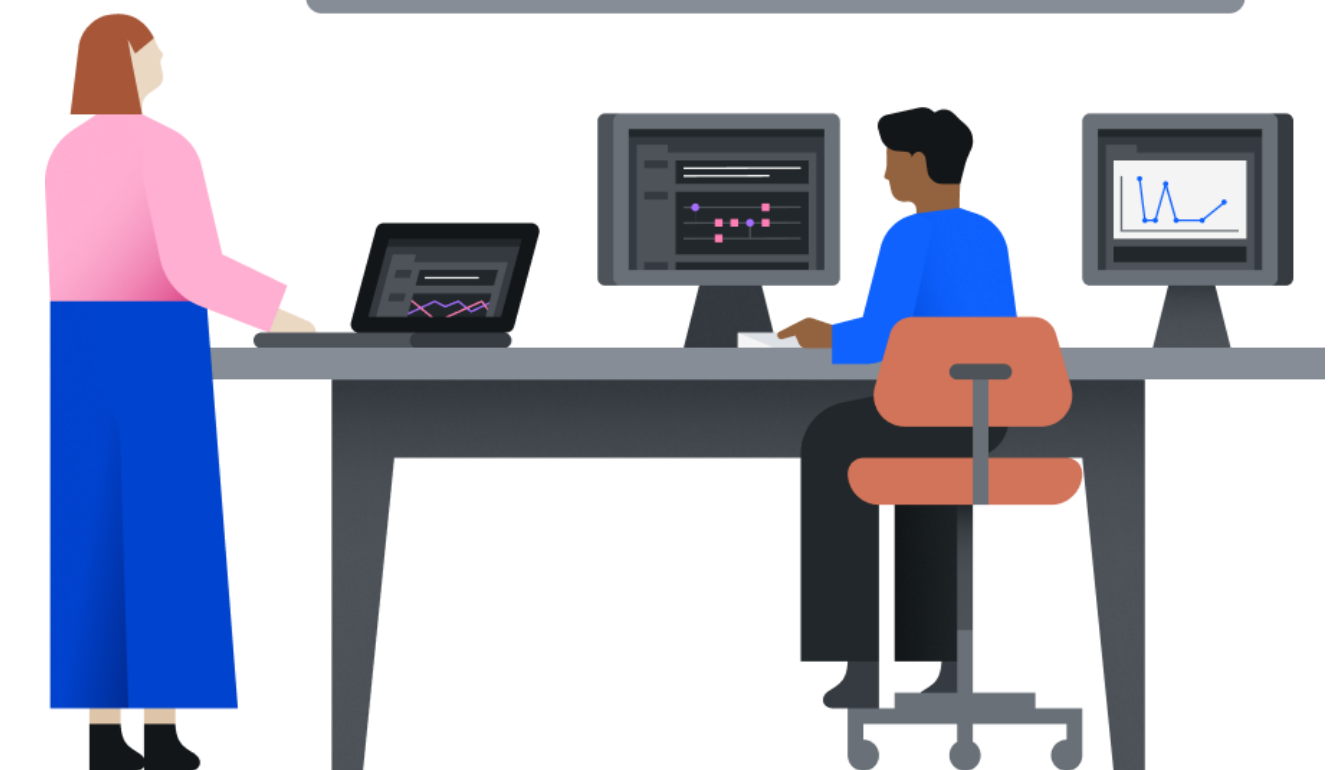
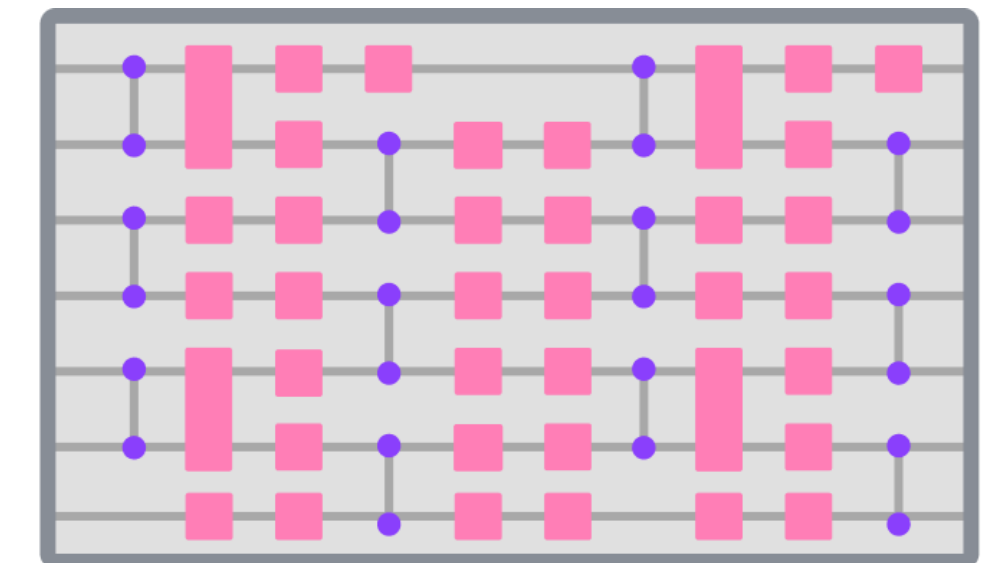
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Our largest educational event every year. Two weeks of lectures, labs, Q&As and community engagement.

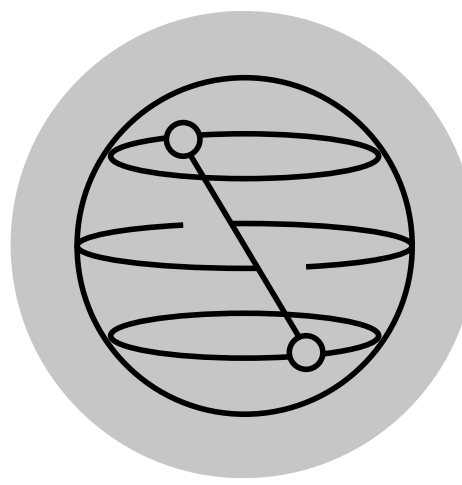
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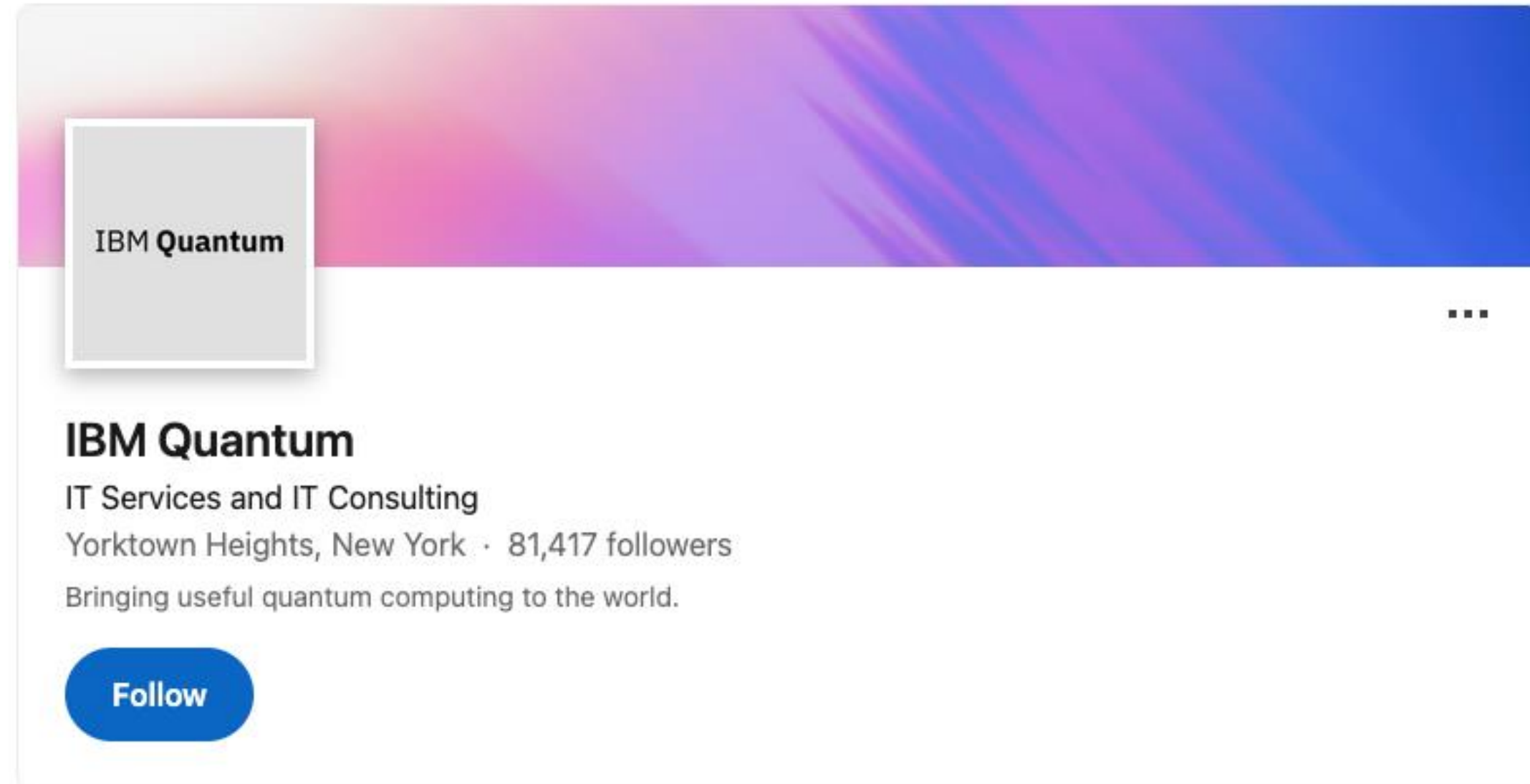


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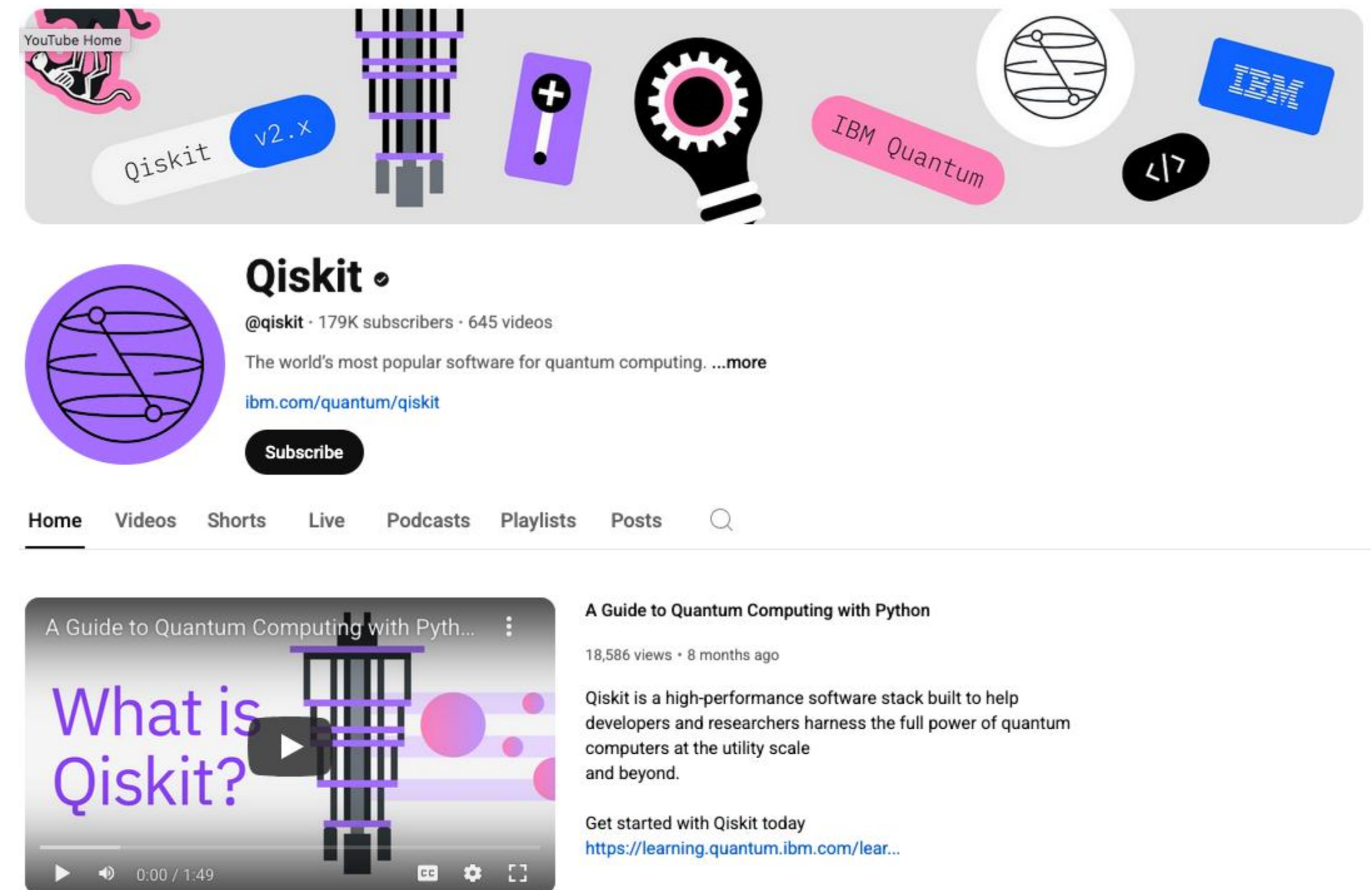
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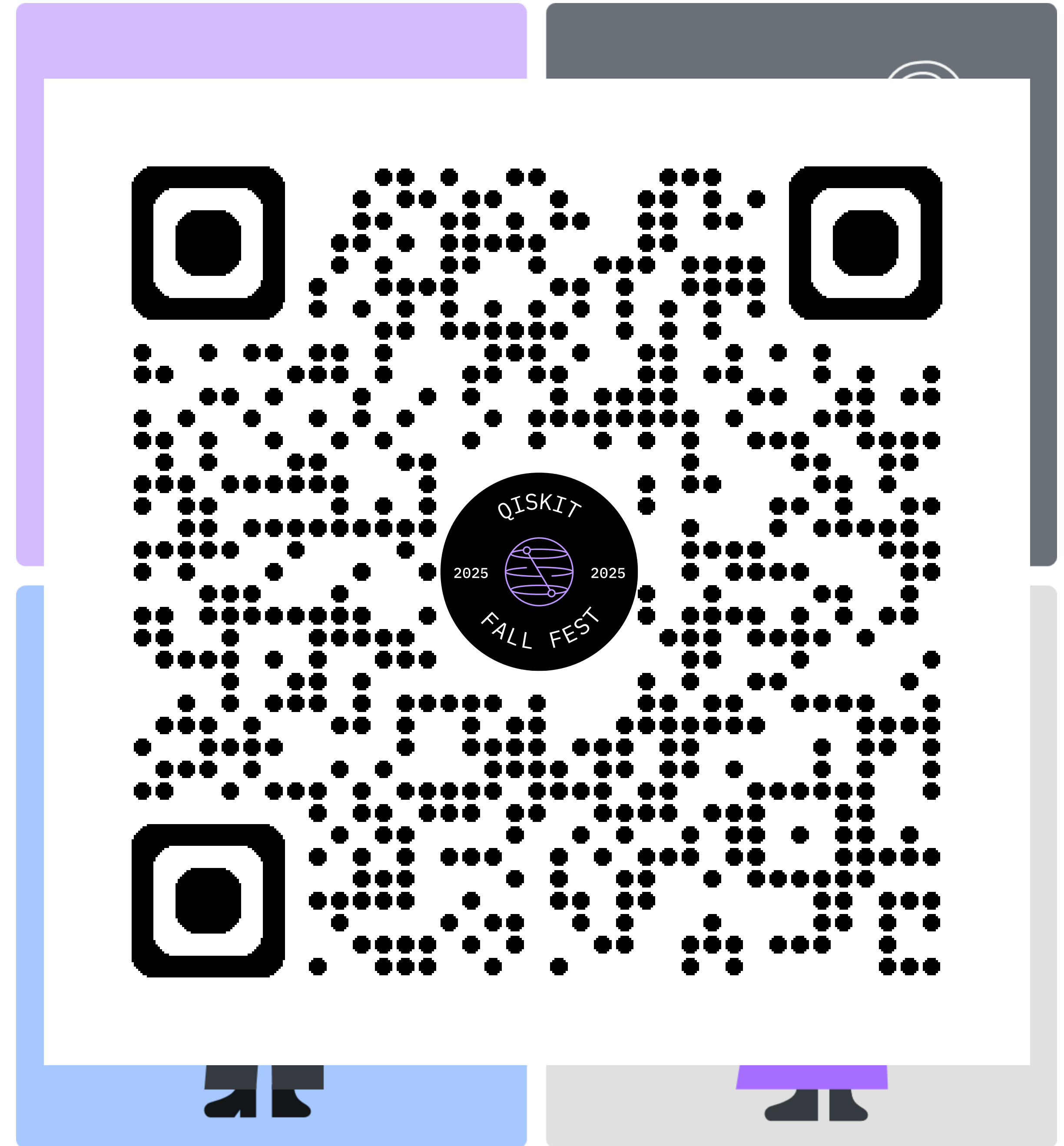


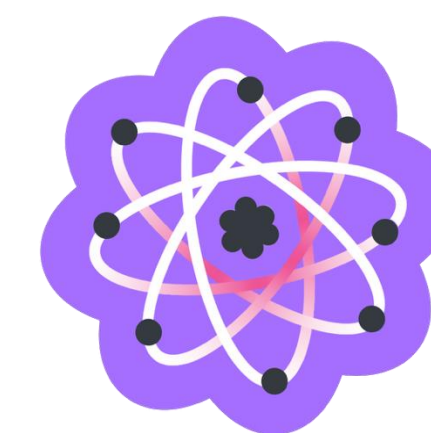
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